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A Systematic Open Source Modeling Method for Energy and Power Systems: Solar PV Application

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ABSTRACT Open source modeling has emerged as a transformative paradigm in energy and power systems studies by enabling transparent, reproducible and collaborative development of models and tools. However, limited access to high-resolution, localized data reduces model accuracy, especially in regions with infrastructural constraints and complex weather conditions. This paper introduces a novel, systematic Open Source Modeling Method for Energy and Power Systems (OSM-MEPS). Derived and synthesized from foundational steps outlined in modeling literature, OSM-MEPS is grounded in first-principles thinking, the scientific method and the engineering design process frameworks, ensuring technical rigor, empirical validation and adaptability. To demonstrate effectiveness, OSM-MEPS guided the modeling and simulation of a solar photovoltaic (PV) system using high-resolution weather and irradiance data for the year 2024 from Solcast-DNV. At the Serres-C site in Greece, OSM-MEPS model results achieved a strong fit with real-world solar PV energy data (MAE: 469.57 kWh, RMSE: 584.62 kWh, R^2 : 0.887), outperforming PVLIB (MAE: 1,400.39 kWh, RMSE: 1,526.11 kWh, R^2 : 0.232). Seasonal testing showed similar results: in May 2024, the OSM-MEPS model achieved R^2 : 0.917 vs. PVLIB's -0.039 ; in December, it maintained R^2 : 0.873 vs. PVLIB's 0.638. Geographic validation demonstrated generalizability. In Pretoria, the OSM-MEPS model outperformed PVLIB (R^2 : 0.443 vs. -0.239); in Adelaide, it maintained accuracy (R^2 : 0.866) while PVLIB failed (R^2 : -0.959). In Kisumu, both models had low correlation despite matching MAE (0.88 kWh), due to data gaps and limited shading metadata. A rooftop energy feasibility study was carried out in Njagu-Mhasibu-Ruiru Estate in Kenya, based on a solar PV installed capacity of 8.75 kW. The OSM-MEPS model predicted an annual yield of 14,329 kWh/year, compared to PVLIB's estimate of 10,048 kWh/year and the GSA reference of 12,784 kWh/year. Overall, the results confirm that the OSM-MEPS method and its resulting model are robust and adaptable tools for high-resolution solar PV energy modeling across diverse geographic regions and weather conditions.

INDEX TERMS Open-source modeling, solar photovoltaic (PV) systems, first-principles thinking, scientific method, engineering design process, rooftop solar PV energy feasibility, model validation.

I. INTRODUCTION

The global energy landscape is fundamentally shifting from fossil fuels to diverse renewables—solar, wind and hydrogen—while the modern power systems grow more complex, requiring integration of variable generation, decentralization and sustainability without compromising

reliability or affordability [1]. However, traditional modeling approaches—often relying on historical trends or analogies to legacy systems—struggle to fully capture the complexity and dynamic nature of emerging challenges such as high renewable penetration, evolving demand patterns and new technologies [2]. To address these limitations, we advocate for a paradigm shift: a systematic energy modeling framework rooted in first-principles thinking. By combining the empirical rigor of the scientific method with the iterative,

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solution-oriented nature of engineering design, this approach fosters resilient, future-ready energy solutions.

First-principles thinking is a problem-solving methodology proposed by Aristotle, defined as “the first basis from which a thing is known” [3], [4]. Also known as first-principles reasoning, this approach involves deconstructing complex systems into their core, fundamental elements, e.g., physical laws, economic constraints, material properties, mathematical principles, consumer needs or human nature, and reconstructing solutions from these foundational truths which is summarized in Figure 1 [4].

The term “foundational” denotes the core principles and concepts (free from assumptions) upon which a system, theory or discipline is built [4]. For example, in energy and power systems, it’s important to question conventional beliefs such as, “centralized grids and lithium-ion batteries are the best options for power systems”. We should ask, “What are the real principles behind how energy is converted, stored and transmitted?” [5]. This approach has led to the discovery of more efficient perovskite solar cells that work in spectrum regions where silicon cells can’t operate [6]. However, perovskite cells still have some challenges, which include stability, cost and durability when exposed to moisture or heat, thus the need for an iterative process for refinement and improvement.

The first-principles thinking approach does not imply making zero assumptions. This approach emphasizes on deep awareness of the foundational assumptions, ensuring they are fundamental, justified and well established within the relevant domain, before building a sound reasoning upon them [5]. It is an iterative process of replacing weaker, unexamined assumptions with stronger, more fundamental ones as knowledge evolves. In other words, assumptions must be justified based on empiricism (measurable evidence) [7], inductive reasoning, and iterative refinement within specific contexts and boundary conditions. Any key assumptions introduced in the modeling process—due to knowledge gaps—should be transparently represented using arbitrary constants, subject to empirical validation [7].

Inventions such as grid-scale gravity-based energy storage offer a promising alternative for short-term grid flexibility. Constructed from recycled materials, these systems function reliably in hot, humid climates, enabling large-scale energy storage without costly batteries [8]. Finally, we need to question the entrenched belief that centralized grid systems are the optimal solution for energy management. Decentralized systems, where energy is generated and distributed locally, offer more flexibility and reliability while reducing the need for expensive transmission infrastructure. The current modeling methodologies frequently neglect such foundational analysis, resulting in solutions constrained by legacy frameworks [9].

For example, consider the challenge of renewable intermittency. Traditional models typically adjust storage capacity based on historical weather patterns. In contrast, a first-

principles approach begins with fundamental relationships—such as the Shockley-Queisser limit [10] governing photovoltaic efficiency and the thermodynamic principles of energy storage—to reimagine system design from its foundations. This methodology has already led to transformative improvements in photovoltaic economics [11]. Similarly, re-evaluating transmission infrastructure through the lens of Ohm’s Law and material conductivity can yield novel designs that minimize energy losses and reduce costs [12].

Open source modeling enables transparent, reproducible and collaborative development of energy models by making the source code, methods and assumptions publicly accessible and modifiable [13]. In OSM-MEPS, the Python code, research findings and underlying physics equations are openly shared (see Appendix). This enables users with high-resolution weather data to adapt and simulate solar PV systems across diverse regions accurately and flexibly, while appropriately citing this work.

This paper introduces a structured method for solar PV energy modeling that explicitly integrates first-principles reasoning with established scientific and engineering frameworks. The scientific method—comprising observation, hypothesis formulation, experimentation and validation—ensures that models remain grounded in empirical constraints, such as the thermodynamic limits of solar energy conversion [14], [15]. Concurrently, the engineering design process—encompassing problem definition, imagination and iterative refinement—translates these principles into feasible solar applications [16]. Applying first-principles thinking throughout the modeling process ensures transparency by grounding decisions in fundamental physical laws. Physical laws are broad, empirically derived principles describing natural phenomena, whereas physics laws are a formal subset—usually expressed mathematically—to explain energy, force, motion and related concepts (e.g., Newton’s laws, Ohm’s law). In this study, both terms are used interchangeably. The OSM-MEPS approach supports clear, rational assumptions and enables site-specific optimization of tilt and azimuth using solar geometry—sun path, surface orientation and irradiance—yielding more accurate PV performance than heuristic methods.

For computational simplicity, this study excluded detailed solar cell circuit modeling, such as the approach in [17]. Instead, the OSM-MEPS method focuses on addressing key gaps in solar PV modeling. It closes theoretical gaps through the scientific method’s iterative hypothesis testing; methodological gaps via first-principles thinking and the engineering design process; and conceptual gaps by unifying objectives, assumptions, calibration and validation into a cohesive framework. Also, OSM-MEPS addresses empirical gaps by quantifying how solar PV orientation (tilt-azimuth) affects energy yield in low and mid-latitude climates. The resulting OSM-MEPS model—developed in Python and fully open-source—is transparent, adaptable and free from closed-box or unclear assumptions.

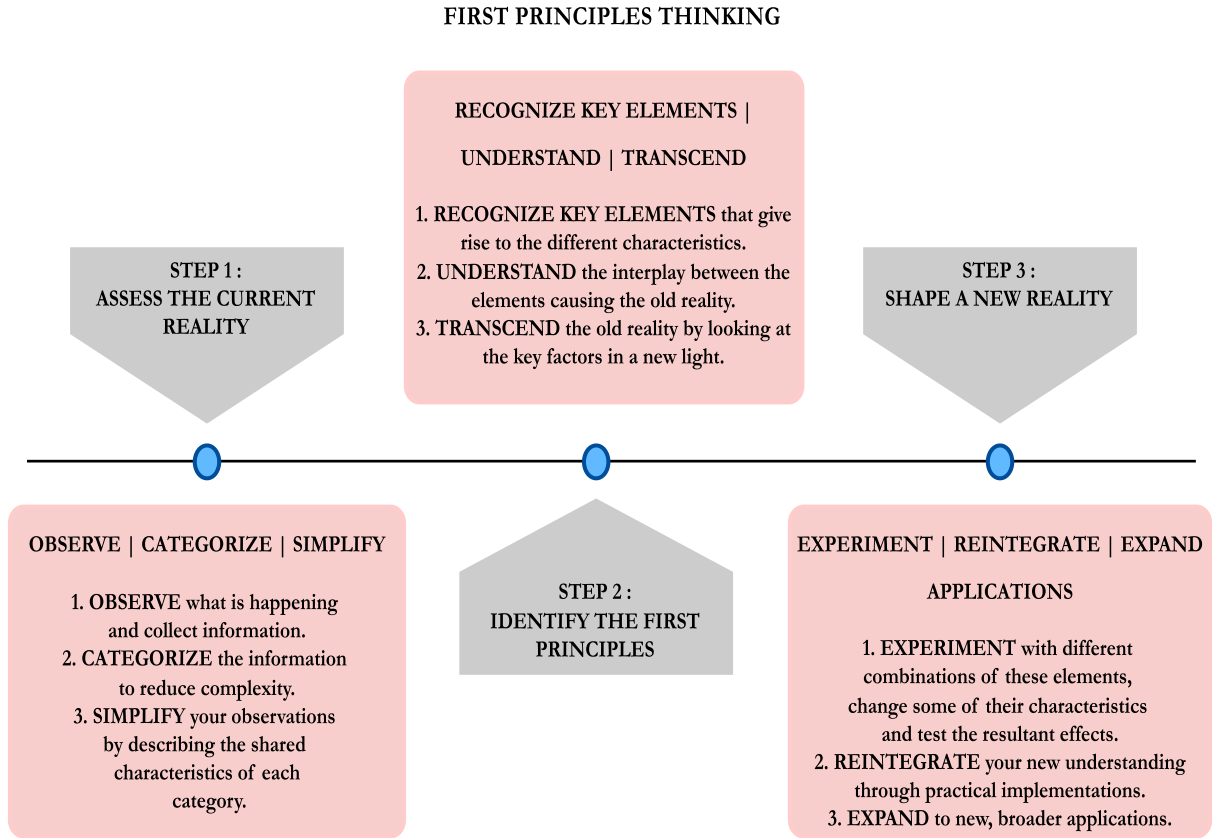


FIGURE 1. First principles thinking [4].

II. LITERATURE REVIEW

A. MODELING OF ENERGY AND POWER SYSTEMS

Energy and power system modeling is indispensable for planning sustainable energy transitions, yet existing approaches face significant limitations that affect their accuracy, scope and policy relevance [2], [18]. These challenges stem from technical, social and methodological constraints, as highlighted by a growing body of literature [2], [18], [19], [20], [21], [22]. Among these, data scarcity remains a foundational barrier. Models require high-quality, granular data to accurately simulate energy demand, solar energy potential and infrastructure performance [2]. However, inconsistencies across regions, fragmented reporting standards and limited access to detailed datasets often lead to oversimplified assumptions [19]. For instance, author [20] emphasizes that incomplete historical datasets for renewable generation or demand profiles force modelers to rely on proxies—risking misrepresentation of real-world variability.

Computational complexity further complicates modeling efforts. High-resolution models that capture grid dynamics—such as transient stability or congestion effects—require substantial computational resources. Author [21] notes that even advanced linear programming techniques often struggle to balance accuracy with tractability, necessitating model simplifications. Author [9] further emphasizes that

such trade-offs may obscure critical interactions between distributed energy resources (DERs) and centralized grids, potentially undermining model insights into real-world system behavior. Uncertainty and risk are inadequately addressed in many frameworks. While scenario-based analyses are common, they frequently overlook low-probability, high-impact events (e.g., geopolitical disruptions or pandemics) [22]. Author [2] argues that probabilistic models often fail to integrate multi-dimensional uncertainties, such as fluctuating technology costs or evolving consumer behavior, leading to overconfident policy recommendations.

The integration of renewable energy sources presents unique challenges. The intermittency of solar and wind generation requires precise modeling of storage and grid flexibility, yet many modeling tools underestimate the temporal resolution needed to capture sub-hourly fluctuations [23]. Furthermore, as author [24] stress, the shift to inverter-based resources introduces stability complexities—such as voltage sag and frequency response—that static models rarely capture. Human and market factors are often oversimplified. Energy models frequently assume rational decision-making by consumers and investors, ignoring cultural preferences or policy resistance [25]. The author [26] illustrates how competition in electricity markets, coupled with regulatory inertia, can distort price signals thus derailing decarbonization pathways.

Resolution constraints affect spatial and temporal granularity. Coarse geographic resolution may overlook rooftop solar potential in urban areas, while hourly aggregation masks demand spikes critical for battery sizing [27]. The author [28] demonstrates how higher-resolution spatial data can alter estimates of the potential of renewable energy by up to 30%. Interdisciplinary gaps hinder holistic analysis. Technical models often neglect environmental externalities, such as land-use conflicts from bioenergy expansion or social equity concerns like energy poverty [29]. Lifecycle impacts—such as rare-earth mineral extraction for batteries—are also frequently excluded, as noted by [30]. Policy and regulatory flexibility is another blind spot. Models calibrated to current regulations may not adapt to abrupt policy shifts, such as carbon tax reforms or renewable portfolio standards [31]. Additionally, geopolitical risks, such as cross-border energy trade disruptions, are rarely modeled dynamically.

Technological evolution introduces further uncertainty. Projections often assume linear cost reductions for technologies like batteries or electrolyzers, underestimating potential breakthroughs [32]. Emerging technologies, such as hydrogen and small modular reactors, remain largely speculative, with ongoing challenges in model validation and calibration. Models trained on historical data may fail to capture future systems dominated by renewables, where weather-dependent generation and demand response introduce new dynamics [33]. The absence of feedback loops between model outputs and actual grid operations further hinders real-world validation.

Additionally, scalability and interoperability issues stem from fragmented modeling ecosystems, as generation, transmission and distribution planning tools often operate in isolation, limiting holistic insights [27]. Despite advocacy from standards bodies such as IEEE and IEC, standardized protocols for integrating distributed energy resources (DERs) are still underutilized.

Physics and technical realism gaps persist. Simplified network models may disregard thermal limits or voltage drops, leading to unrealistic congestion estimates [21]. Nonlinear phenomena, e.g., cascading failures, are often linearized for computational ease.

Addressing these limitations requires advances in data collection techniques, accurate modeling and interdisciplinary approaches. Emerging techniques such as machine learning and agent-based modeling offer promise for capturing complexity and uncertainty [33], however, as the author [31] emphasizes, transparency in modeling assumptions is vital to ensure that models guide equitable and resilient energy transitions.

B. AN OVERVIEW OF THE SCIENTIFIC METHOD

- 1) **Question:** The process begins with a question or observation about a phenomenon that sparks curiosity or attention. This question sets the direction for the research [34], [35].

- 2) **Background Research:** Before formulating a hypothesis, researchers conduct background research to gather existing knowledge related to the question. This helps in understanding the context and informs the development of a hypothesis [34], [35].
- 3) **Construct a Hypothesis:** Based on the background research, a hypothesis is formulated. This is a testable statement predicting the relationship between variables. It serves as a foundation for further experimentation [34], [35].
- 4) **Experiment:** Researchers design and conduct experiments to test the hypothesis. This involves defining procedures, selecting materials and establishing controls to ensure valid results [34], [35].
- 5) **Procedure Working?** After conducting the experiment, researchers evaluate whether the procedure worked as intended. If not, troubleshooting is necessary to identify and resolve issues in the experimental design or execution [34], [35].
- 6) **Data Analysis and Conclusion:** Once the experiment is completed, data is collected and analyzed to determine if it supports or contradicts the hypothesis. Statistical methods may be employed to assess significance [34], [35].
- 7) **Results Align with Hypothesis:** If the results align with the hypothesis, this supports its validity. However, if results only partially align or do not align at all, this indicates that further investigation is needed [34], [35].
- 8) **Communicate Results:** Regardless of the outcome, researchers must communicate their findings through reports, publications or presentations. Sharing results contributes to collective knowledge and allows others to learn from the research [34], [35].
- 9) **Troubleshoot Procedure:** If issues arose during experimentation (e.g., unexpected results), researchers revisit their procedures to troubleshoot and refine their approach for future experiments [34], [35].
- 10) **Iterate:** The scientific method is iterative; findings from one project can inform future research questions and hypotheses. Data collected can serve as background research for subsequent studies [34], [35].

The scientific method is essential for conducting rigorous scientific inquiries [34], [35]. It provides a structured framework that encourages critical thinking and systematic investigation of phenomena [34], [35]. This iterative process, as shown in Figure 4 ensures continuous improvement and adaptation in scientific research, ultimately advancing our understanding of complex systems and relationships in nature [34], [35].

Meteorology and climate science play a crucial role in energy systems modeling, as shown in Table 1. GHI data from [36] and [37] Solargis as presented in the Figure 2 heat map reveals that tropical regions, especially deserts and arid zones, have higher solar potential than the equator, as shown in Figure 3. The solar insolation theory (SIT) suggests that

the equator receives the highest solar energy due to the direct angle of sunlight [38]. In contrast, higher latitudes receive less due to the oblique angle of sunlight [38].

However, real-world data from sources such as Solargis show that certain tropical regions (just outside the equator) can have higher annual Global Horizontal Irradiance (GHI) than the equator. According to SIT, we expect the equator to have the highest GHI due to its consistent sun angles year-round [38] but data and modeling results show that factors like cloud cover and atmospheric humidity attenuate solar radiation. The Intertropical Convergence Zone (ITCZ) [39], [40], which spans much of the equator, causes frequent cloud cover, leading to scattered and diffuse radiation, significantly impacting the effective GHI [41].

TABLE 1. GHI in Equatorial vs tropical regions [36], [37].

Factors	Equatorial Region	Tropical Region
Cloud Cover Patterns	High cloud cover due to ITCZ, e.g., Congo Basin (Central Africa) and Amazon Rainforest (South America).	Clearer skies in dry regions, e.g., Sahel (Africa) and Namib Desert (Namibia).
Seasonal Variability	Minimal seasonal changes but persistent clouds, e.g., Kampala, Uganda and Singapore.	Distinct dry seasons with clear skies, e.g., Windhoek, Namibia and Cairns, Australia.
Atmospheric Effects	High water vapor levels reduce GHI, e.g., Jakarta, Indonesia and Manaus, Brazil.	Low water vapor in arid regions enhances GHI, e.g., Khartoum, Sudan and Windhoek, Namibia.
Terrain and Surface Albedo	Dense vegetation absorbs sunlight, e.g., Equatorial Guinea and Brunei.	High albedo in deserts boosts reflected sunlight, e.g., Namib Desert (Namibia) and Atacama Desert (Chile).
Diffuse Irradiance Contribution	Persistent dense clouds reduce both direct and diffuse sunlight, e.g., Libreville, Gabon and Medellín, Colombia.	Scattered clouds increase diffuse sunlight, e.g., Darwin, Australia and Dakar, Senegal.
Horizon and Sun Path	Near-vertical sun paths year-round, but dense forests or frequent clouds obstruct sunlight, e.g., Kisangani, DRC and Quito, Ecuador.	Slightly angled sun paths with clear horizons boost GHI, e.g., Windhoek, Namibia and Lusaka, Zambia.

C. AN OVERVIEW OF THE ENGINEERING DESIGN PROCESS

The Engineering Design Process, as shown in Figure 5, can be summarized as follows:

- 1) **Ask and Define the Problem:** The initial step involves clearly identifying the problem or need the design aims to address [42], [43].
- 2) **Do Background Research and Specify Requirements:** Conduct thorough research to understand existing solutions, relevant technologies and specific

design criteria. This step includes defining the constraints and performance requirements. It also helps avoid potential pitfalls [42], [43].

- 3) **Imagine, Brainstorm Ideas, Choose a Feasible Idea:** Generate a range of potential solutions through brainstorming. Evaluate these ideas and select the most promising and feasible one [42], [43].
- 4) **Evaluate and Plan:** Develop a detailed plan outlining the steps required to create and test the design. This includes resource allocation and timeline considerations [42], [43].
- 5) **Develop/Create Your Design/Model:** Implement the design plan by creating a prototype or model of the proposed solution [42], [43].
- 6) **Test Performance:** Evaluate the performance of the prototype or model against the specified requirements [42], [43].
- 7) **Solution Meets Requirements?** Determine if the initial testing results meet the defined criteria. If met, continue to step 10 [42], [43].
- 8) **Solution Meets Requirements Partially or Not at All?** Based on testing results, identify areas for improvement or alternative solutions [42], [43].
- 9) **Improve Based on Testing:** Refine the design based on the testing results. This iterative process continues until the solution meets the desired requirements [42], [43].
- 10) **Communicate Findings:** Document and share the design process, results and lessons learned [42], [43].

D. AN OVERVIEW OF THE KEY STEPS IN ENERGY AND POWER SYSTEMS MODELING

The energy and power systems modeling process presented in this section involves a series of critical steps, consolidated from scholarly literature and methodological frameworks [44], [45]. *Step 1:* Initiate the modeling effort by clearly defining the **scope and objectives** [46], including specific goals such as enhancing energy security, minimizing costs or integrating renewable energy sources. Then, determine the geographical boundaries (e.g., local, national or regional) and identify the relevant sectors or technologies to be examined, such as power generation, transportation or specific options like solar PV energy [46]. *Step 2:* Perform comprehensive **data collection and preparation**, which directly influences the outcomes and reliability of the analysis [18], [47], [48], [49], [50]. Data collection covers several key categories [48].

1) HISTORICAL ENERGY CONSUMPTION PATTERNS

One of the first data sets to gather is historical energy consumption data. This data is necessary to understand demand fluctuations, peak usage times and seasonal variations. It helps predict future energy demand by identifying trends and establishing reliable consumption models [51].

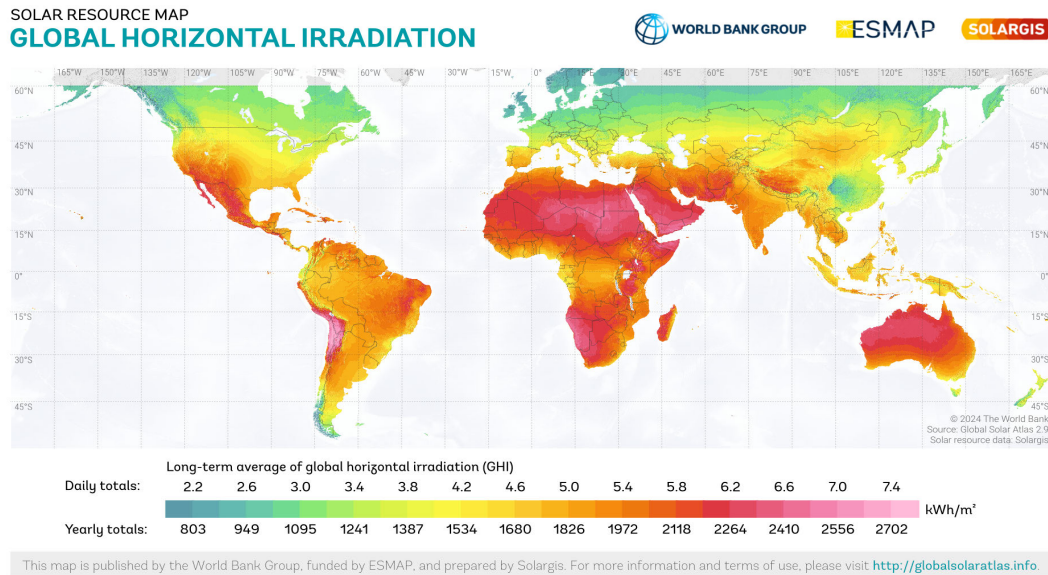


FIGURE 2. Long-term (10-year) averaged Global Horizontal Irradiance (GHI) map. [37].

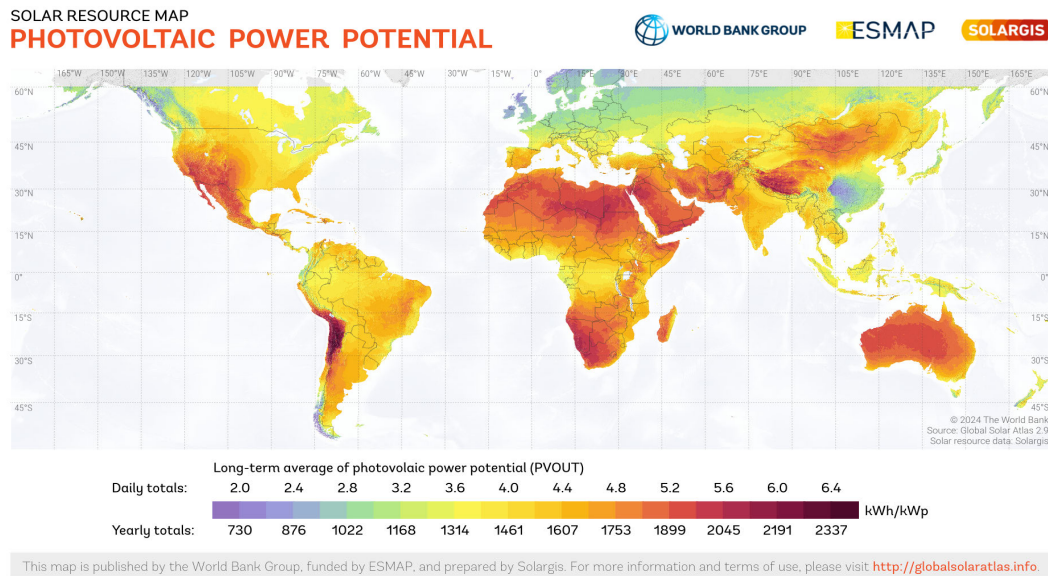


FIGURE 3. Long-term (10-year) averaged photovoltaic power potential. [37].

2) GENERATION CAPACITIES

Data on power plants' generation capacities, technology, efficiency rates, and operational statuses is essential for accurately assessing energy supply and optimizing system integration [47], [51].

3) ECONOMIC INDICATORS

Understanding the broader economic context is equally essential in energy systems modeling [52], [53]. Macroeconomic indicators—such as GDP growth, industrial output and population trends—influence energy consumption patterns and support more accurate projections of future demand [53], [54], [55].

4) WEATHER DATA FOR RENEWABLE ENERGY POTENTIAL

Accurate weather data—such as solar irradiance, relative humidity, wind speed and rainfall—is essential for modeling renewable energy production and forecasting generation profiles for solar, wind and hydropower, thereby supporting system stability [48], [56].

5) GROUND SURVEYS AND SCADA SYSTEMS

High-fidelity data from ground surveys and real-time SCADA systems—such as voltage, frequency and generation output—is invaluable for refining models and simulating grid dynamics [57], [58]. To ensure reliable simulations and informed grid-scale energy decisions, modelers integrate

THE SCIENTIFIC METHOD

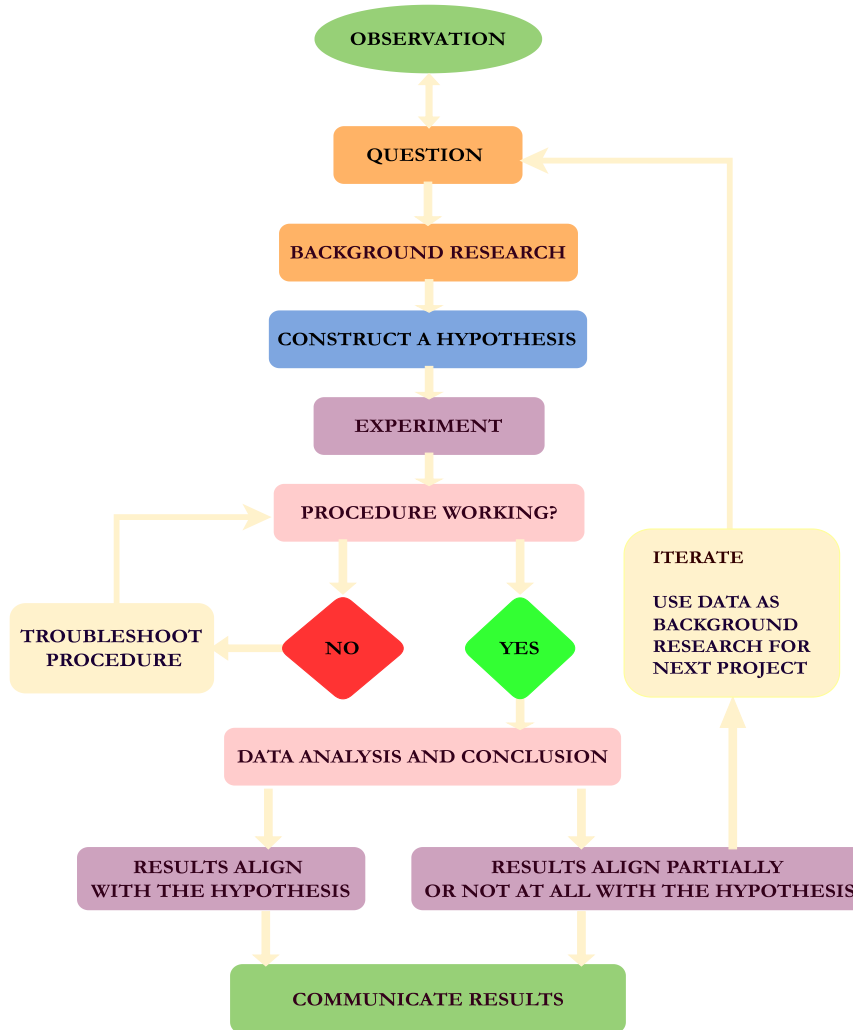


FIGURE 4. The scientific method [34], [35].

cleaned and verified data from storage, demand response and the broader power system, emphasizing accuracy and system-wide completeness.

Next, in *step 3*: Choose a **suitable modeling framework**. A framework is a structured method that simulates and optimizes energy systems using collected data. Select a framework designed to meet specific objectives, e.g., finding cost-effective energy solutions or assessing policy impacts [59]. Examples include PyPSA, PandaPower, Open Energy Modeling Framework (oemof) and PVLIB. Align your framework choice with your objectives to ensure proper model use.

After selecting the framework, proceed to *step 4*: and formulate the model. **Model formulation** involves creating a mathematical or computational representation of a system, phenomenon or process [60]. This step converts real-world concepts and constraints into a structured format for analysis. It typically includes defining variables, relationships and

equations that describe how different system elements interact [60].

Model formulation in energy and power systems entails defining system components (e.g., generators, transmission lines), decision variables (e.g., generation capacities) and an objective function (e.g., cost minimization), along with constraints such as supply-demand balance and environmental limits. This structured approach enables optimization and simulation, supporting informed decisions on system design and operation. The choice of formulation depends on the model type, aiming to balance accuracy with computational efficiency [61], [62].

After formulating the model, *step 5*: **implement** it within the chosen computational framework using optimization software, programming languages (e.g., Python) or specialized modeling platforms [60]. Translate the model's logic and equations into executable code and perform preliminary tests to verify correct functionality. These tests help detect

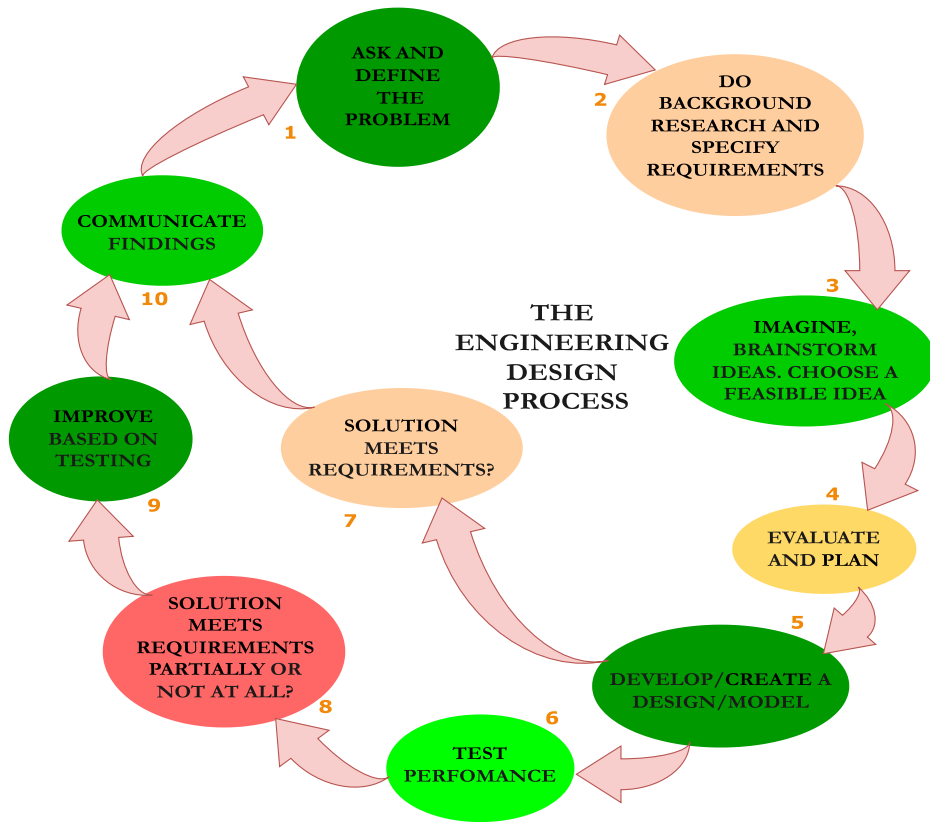


FIGURE 5. Engineering design process [42], [43].

and resolve any formulation issues before proceeding to large-scale simulations or real-world applications [60].

In *Step 6: Calibrate and validate* the model to ensure accuracy and reliability by adjusting its parameters based on historical data—such as past energy demand and weather conditions—to enhance predictive performance [48], [63]. After calibration, validation assesses the model’s ability to predict outcomes using new, unseen data [56]. Strong validation performance confirms the model’s robustness and suitability for applications such as dynamic simulations under extreme conditions [48], [56]. This step ensures the model is not only fitted to historical data but can also reliably forecast future or unfamiliar scenarios [27].

After validating the model, *step 7: develop scenarios*. Create hypothetical future conditions to test how the energy system performs under different circumstances [61], [64]. For example, policymakers might explore the impacts of carbon pricing or renewable energy subsidies [48], [56]. Technological advancements—such as the adoption of electric vehicles—can significantly impact both energy demand and supply. Additionally, energy prices influence transmission system congestion and operational performance [65]. Through scenario development, modelers can evaluate system resilience, identify potential vulnerabilities and support informed decisions regarding infrastructure investments and policy [44], [66]. Scenario analysis incorporates uncertainties

and trade-offs, enabling the energy system to be designed and adapted for future challenges [48], [56].

Next, *step 8: analyze and interpret the results* to extract actionable insights. Assess the system’s reliability, sustainability and resilience under various scenarios to evaluate its adaptability [48], [63]. Clearly communicate findings to stakeholders—such as policymakers and the public—through well-documented methodologies, transparent assumptions and accessible visualizations (e.g., graphs and charts). This ensures that evidence-based conclusions are both understandable and useful for informed decision-making [65].

To continuously improve energy models, actively engage in *step 9: the review and conclusion* stage. Gather feedback from stakeholders, such as industry experts and utilities, to evaluate the model’s effectiveness and relevance [59]. Stakeholders help identify potential gaps, assumptions needing revision or new developments for modeling iterations. *Step 10: effective communication* plays a vital role in sharing key insights from different modeling scenarios with stakeholders. It also helps restart the feedback loop, allowing new data, such as updates in technology, market trends, or regulations, to be integrated into the model. This ongoing iteration strengthens the model’s robustness and supports more informed decision-making in the face of complex and uncertain energy systems [59].

III. METHODOLOGY

A. THE SYSTEMATIC OPEN SOURCE MODELING METHOD FOR ENERGY AND POWER SYSTEMS (OSM-MEPS)

This section presents the OSM-MEPS method as shown in Figure 6. The **ten key procedural steps** are synthesized and derived from existing literature. When adapted to a particular modeling of power or energy systems, OSM-MEPS aims to systematically address the challenges highlighted in the literature review section. However, in this study, OSM-MEPS rigor and robustness are demonstrated through solar photovoltaic (PV) energy and power modeling to provide validation, transparency and reproducibility. This method follows a structured, iterative workflow that integrates first-principles thinking, the scientific method and the engineering design process aspects, tailored to the specific modeling context and technical requirements:

1) ADDRESSING DATA SCARCITY

- **Step 2 (Data Collection and Preparation)** enforces rigorous data collection and preprocessing, prioritizing high-quality and granular key weather inputs.
- **First-Principles Thinking** (Observe, Categorize, Simplify) encourages modelers to identify fundamental variables and relationships, minimizing assumptions where data gaps exist [4].
- **Calibration and Validation (Step 6)** ensures empirical-based adjustments to solar PV models, improving accuracy even with imperfect datasets.

2) MITIGATING COMPUTATIONAL COMPLEXITY

- **Step 3: Modeling framework selection** recommends open-source tools such as Python for programming and Jupyter Notebooks for interactive development and visualization. Python's flexibility, extensive libraries (e.g., SciPy for numerical methods, Dask for parallel computing) and efficient handling of complex, high-resolution models make it a preferred choice in energy system modeling and simulation.
- **Step 4: Model Formulation:** Solar PV modelers incorporate weather data—such as GHI, temperature, wind speed, and cloud cover—to calculate effective irradiance and module temperature. Performance is benchmarked against Standard Test Conditions (STC) and adjusted using temperature coefficients and cell temperature models (e.g., NOCT) to reflect actual operating conditions.
- **Iterative Refinement** improves computational efficiency by selectively adding complexity to critical components, avoiding unnecessary detail based on context. This approach balances accuracy and computational cost by focusing resources where they matter most.

TABLE 2. Framework-based mapping of energy and power modeling steps.

Key Method Step	Framework Integration
Observation and Problem Definition	Scientific Method: Collect high-resolution, fine-granular empirical data (e.g., Solcast's 5-minute DNI, GHI, temperature, cloud opacity). Engineering Design: Define practical goals (e.g., "Assess rooftop solar PV energy potential in Kenya"). First-Principles: Identify core physics principles (e.g., solar irradiance equations).
Hypothesis and Model Development	Scientific Method: Formulate hypotheses (e.g., "High-resolution weather data improves accuracy" or "The alteration of the PV array azimuth angle has a minimal effect on energy output in equatorial regions when the tilt angle is fixed at low values across all seasons."). First-Principles: Derive equations (e.g., $\text{energy yield} = \text{irradiance} \times \text{area} \times \text{efficiency}$). Engineering Design: Develop modular components (solar PV performance model, module temperature model, etc.).
Experimentation and Calibration	Scientific Method: Test models against benchmarks (e.g., Serres C PV plant, PVLIB, GSA solargis, SAM). First-Principles: Adjust parameters using fundamental physics (e.g., Simulate the effects of a PV array's azimuth angle on output energy using a model based on physical principles). Engineering Design: Iterate through solutions to refine the model.
Analysis and Iteration	Scientific Method: Compare outputs to empirical data (e.g., OSM-MEPS results vs Serres C PV plant energy output). Engineering Design: Refine solutions (e.g., model shading losses, tune efficiency to cover for any unmodeled losses or upgrade in inverter technology scenarios, e.g., 98% efficiency). First-Principles: Ensure equations remain consistent with physics principles during system scaling.

3) MANAGING UNCERTAINTY

- **Step 7: (Scenario Development)** in photovoltaic (PV) system design involves simulating various configurations to evaluate performance under different conditions. A key aspect of this process is analyzing how different tilt and azimuth angles affect energy generation. Simulating scenarios with varying tilt and azimuth angles helps identify the optimal configuration for maximum solar energy yield, especially when accounting for seasonal changes and geographic location. This approach supports informed decision-making in solar PV deployment.
- **The Scientific Method** involves hypothesis testing through simulations and comparison with real-world data, while the **Engineering Design** aspect emphasizes iterative prototyping and validation to refine model accuracy.

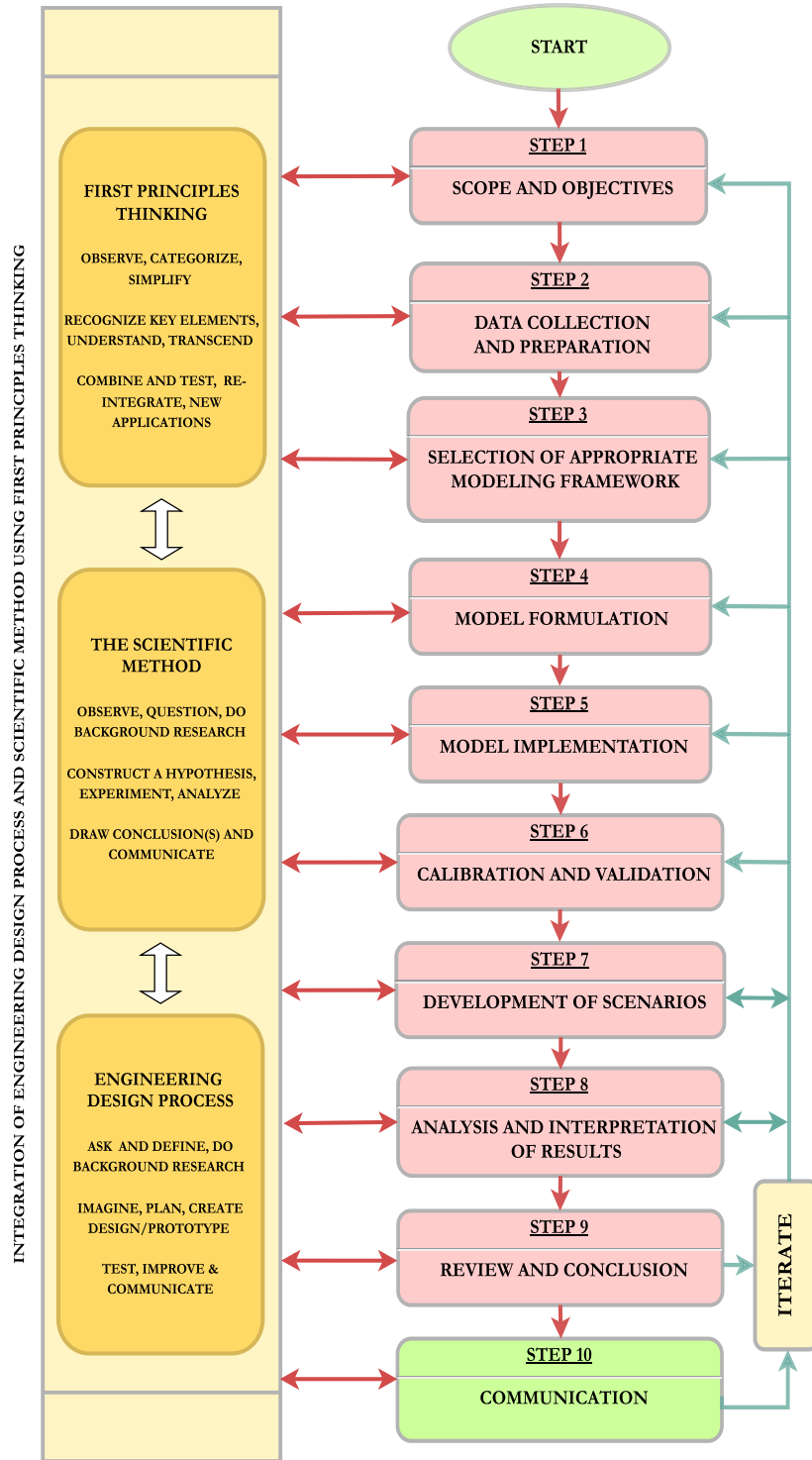


FIGURE 6. A systematic open-source modeling method for energy and power systems (OSM-MEPS).

IV. CASE STUDY: GREECE, SOUTH AFRICA, AUSTRALIA AND KENYA

A. MODELING AND VALIDATION OF SOLAR PV ENERGY SYSTEM USING OSM-MEPS METHOD

DNI (Direct Normal Irradiance) is the solar radiation received perpendicularly from the sun's direct beam,

excluding diffuse or reflected light. **DHI** (Diffuse Horizontal Irradiance) is the scattered solar radiation reaching a horizontal surface from the sky, excluding the sun's direct beam and dominates under cloudy conditions. **GHI** (Global Horizontal Irradiance) combines DHI with the DNI: $GHI = DNI \cdot \cos(\theta_z) + DHI$, where θ_z is the solar zenith angle.

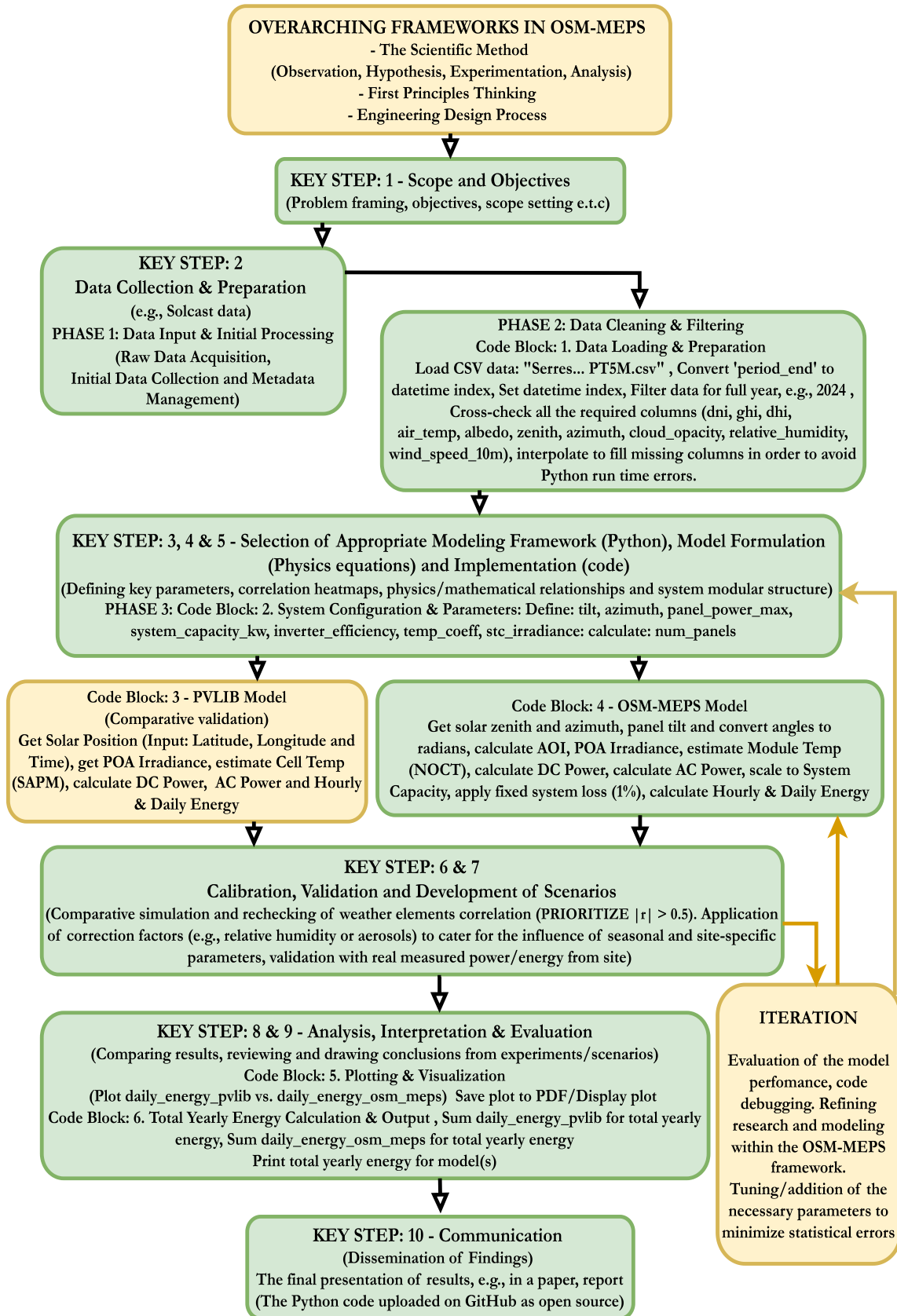


FIGURE 7. OSM-MEPS workflow and implementation flowchart: Solar PV power and energy modeling.

GTI (*Global Tilted Irradiance*) or **Plane-of-Array (POA)** irradiance is the solar radiation incident on a tilted surface. **Albedo** (β) is the ground reflectivity, ranging from 0.1 (dark soil) to 0.95 (fresh snow), contributing to GTI via reflected irradiance, especially for downward-facing arrays [67], [68], [69].

In solar irradiance modeling, the **Angle of Incidence (AOI)** is the angle between the solar ray and the panel's surface normal, influencing how much direct solar energy is absorbed. Solar tracking angles—tilt (panel's elevation from horizontal) and azimuth (compass direction the panel faces, with 0° being the true north)—are used to optimize sunlight exposure. In tilted panels, sky diffuse irradiance refers to the portion of scattered sunlight that comes directly from the sky dome, excluding light reflected from the ground (albedo) [67], [68], [69].

High-resolution, validated key weather data was obtained from Solcast, a DNV company, at a 5-minute temporal resolution. The results of the OSM-MEPS solar PV energy model, factoring PV and inverter efficiency, for Serres-C 24 are compared to the PVLIB's model [67], [68], [69] and real-world data from the Serres-C solar power plant energy output (kWh) as shown in Figures 14 and 15. This is a key step in the validation process. OSM-MEPS ensures that models are first validated using empirical data before being applied to data-scarce sites, enhancing credibility and generalization accuracy. The System Advisor Model (SAM) and Global Solar Atlas (GSA) annual energy graphs—based on their default 60-minute weather data—were plotted separately to avoid resolution mismatches. Seasonal trends and total energy outputs are shown in both graph and table form, aligning closely with OSM-MEPS model results.

Correlation heatmaps provide a visual representation of the relationships between variables, making them a valuable tool for analyzing weather data and optimizing energy systems. This is a crucial step in the procedure shown in Figure 7. This paper examines two heatmaps for the year 2024, focusing on Njagu-Mhasibu-Ruiru Estate, Kenya, and Serres C Solar Power Plant, Greece as presented in the heatmaps in Figures 9 and 10. The heatmap interpretation for other select regions as depicted in Figures 11, 12, 13 follows the general interpretation as stated in subsection IV-B. The analysis identifies general patterns, key relationships, and location-specific differences, offering insights for energy modeling and decision-making.

B. GENERAL INTERPRETATION OF CORRELATION HEATMAPS

Correlation coefficients range from -1 to $+1$, where $+1$ indicates perfect positive correlation, 0.0 (zero) means no correlation and -1 indicates perfect negative correlation. In the heatmaps, color intensity reflects the strength of the relationship, with darker hues denoting stronger correlations. The diagonal entries (1.0) represent self-correlation.

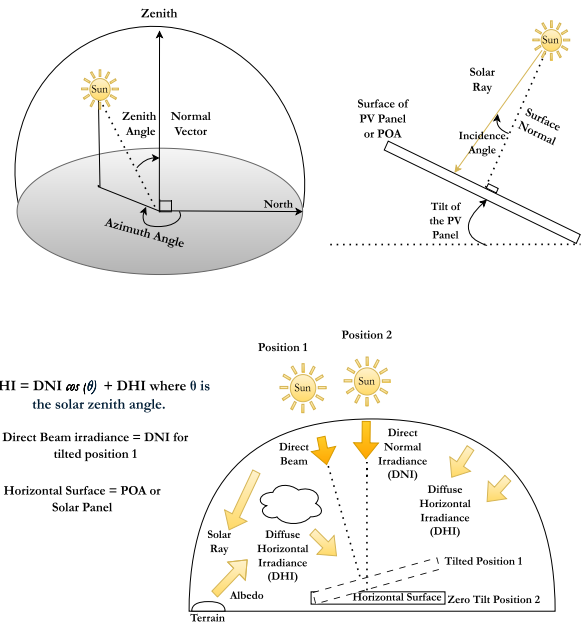


FIGURE 8. Solar tracking angles and irradiance model [70].

Key observations for Serres C include strong correlations among clearsky irradiance variables (0.80 – 0.98) and between GHI/GTI (0.98). The zenith angle exhibits strong negative correlations with clear-sky metrics (-0.84 to -0.88), consistent with solar geometry. Unlike Kenya, albedo correlations are weak, likely due to regional surface reflectivity differences. Similar to Serres C, clearsky metrics in Kenya show strong internal correlations (0.84 – 0.98). Relative humidity negatively correlates with irradiance (-0.65), highlighting cloud-cover impacts. It's crucial to ensure data quality and representativeness. Correlation does not imply causation; further analysis is needed to establish causal relationships. This is achieved by comprehensively and iteratively modeling using first principles and validating the system with site-measured Solar PV energy output data.

C. PHYSICS MODELING USING OSM-MEPS METHOD: FIRST PRINCIPLES APPROACH

This section explains how OSM-MEPS applies the first principles thinking to outperform the PVLIB and SAM model equations and approach in the context of solar PV system energy quantification.

1) EXPLICIT GEOMETRIC ANGLE-OF-INCIDENCE (AOI) CALCULATION

The solar angle of incidence (AOI) is directly computed using spherical trigonometry, based on first principles [71]. The key variables—solar zenith angle, solar azimuth, panel tilt and panel azimuth—are all angular positions on a conceptual celestial sphere, allowing for precise geometric modeling of

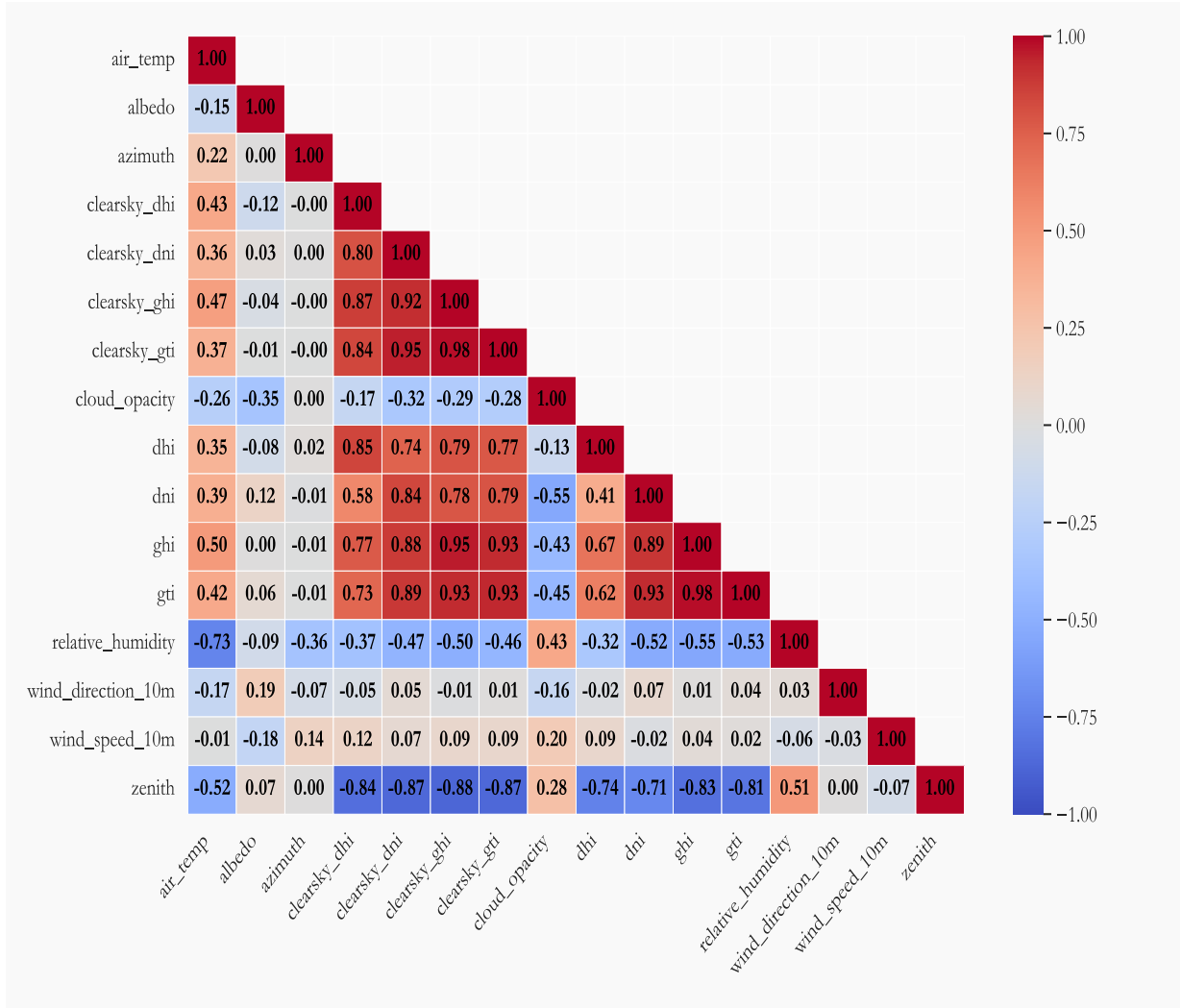


FIGURE 9. Correlation of weather elements at Serres-C, Greece.

the sun's position relative to the panel. The resultant equation is as follows:

$$\theta_{AOI} = \cos^{-1} \left(\cos(\theta_{zenith}) \cos(\theta_{tilt}) + \sin(\theta_{zenith}) \sin(\theta_{tilt}) + \cos(\theta_{sun\ azimuth} - \theta_{panel\ azimuth}) \right) \quad (1)$$

where θ_{zenith} is the solar zenith angle, θ_{tilt} is the panel tilt, $\theta_{sun\ azimuth}$ is the solar azimuth, and $\theta_{panel\ azimuth}$ is the panel azimuth. This approach yields a more accurate plane-of-array (POA) total irradiance than PVLIB's, primarily due to the higher quality input data from Solcast, which enhances the modeling of shading and cloud opacity effects. While PVLIB can utilize the National Renewable Energy Laboratory's Solar Position Algorithm (SPA) for solar position calculations—renowned for its precision—the SPA involves over 47 detailed astronomical steps and corrections [72], which adds complexity and increases computation time.

2) INCORPORATION OF CLOUD OPACITY AS AN EXPLICIT MODIFIER

The model explicitly reduces direct irradiance by the cloud opacity fraction:

$$I_{poa, direct} = DNI \times \cos(\theta_{AOI}) \times \left(1 - \frac{\text{cloud opacity}}{100} \right) \quad (2)$$

This physically grounded correction accounts for atmospheric attenuation beyond basic diffuse irradiance, enhancing realism under variable weather conditions. By incorporating local empirical adjustments, it refines Solcast's generalized data—improving accuracy for site-specific conditions or microclimates, especially where ground-based validation stations are absent in the Solcast model.

3) SEPARATE DIFFUSE AND ALBEDO COMPONENTS WITH GEOMETRY

Diffuse irradiance is split into two components:

- Sky diffuse: scaled by $\frac{1+\cos(\theta_{tilt})}{2}$

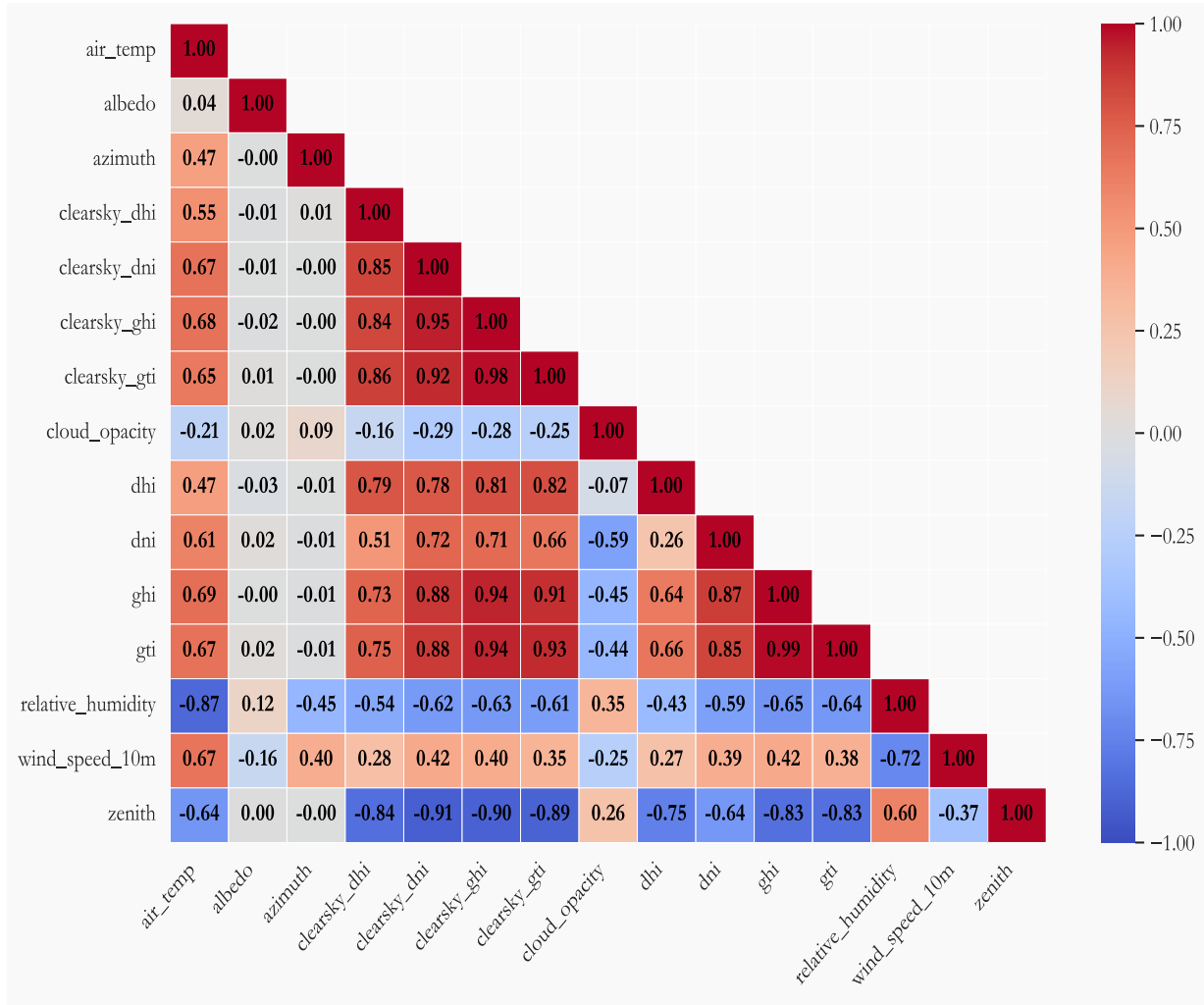


FIGURE 10. Correlation of weather elements at Njagu-Mhasibu-Ruiru Estate, Kenya.

- Ground reflected (albedo): scaled by $\frac{1-\cos(\theta_{\text{tilt}})}{2} \times \text{albedo}$

The ground-reflected (albedo) irradiance $I_{\text{poa,diffuse}}$ is calculated by multiplying the GHI by the above scaling coefficient. Since GHI includes both direct and diffuse solar radiation reaching the ground, it provides a suitable basis for estimating the portion that reflects from the surrounding environment. This component adds to the total irradiance received by tilted PV panels. The approach is grounded in fundamental solar irradiance principles.

4) MODULE TEMPERATURE ESTIMATION

The module temperature is modeled as:

$$T_{\text{module}} = T_{\text{NOCT}} + \frac{I_{\text{poa,total}}}{1000} (28 - T_{\text{air}}) \quad (3)$$

This captures thermal effects of irradiance and ambient temperature on cell efficiency more directly than empirical temperature models, improving power output estimates.

5) TEMPERATURE AND HUMIDITY CORRECTION OF POWER OUTPUT

The DC power output includes corrections for environmental factors:

$$P_{\text{panel}} = P_{\text{panel}}^{\text{max}} (1 + \alpha_T (T_{\text{module}} - T_{\text{NOCT}})) \quad (4)$$

$$P_{\text{dc}} = P_{\text{panel}} \frac{I_{\text{poa,total}}}{I_{\text{STC}}} \quad (5)$$

$$P_{\text{dc,hum}} = P_{\text{dc}} (1 - k_{\text{RH}} \times RH) \quad (6)$$

where $P_{\text{panel}}^{\text{max}}$ is the maximum power rating of the solar panel under standard test conditions (STC), α_T is the temperature coefficient of power (typically negative, indicating power decreases with increasing temperature), T_{module} is the operating temperature of the solar module in degrees celsius and T_{NOCT} is the nominal operating cell temperature, usually around 45 °C. The term $I_{\text{poa,total}}$ represents the total plane-of-array irradiance incident on the solar panel surface (in W/m²), while I_{STC} is the reference irradiance at

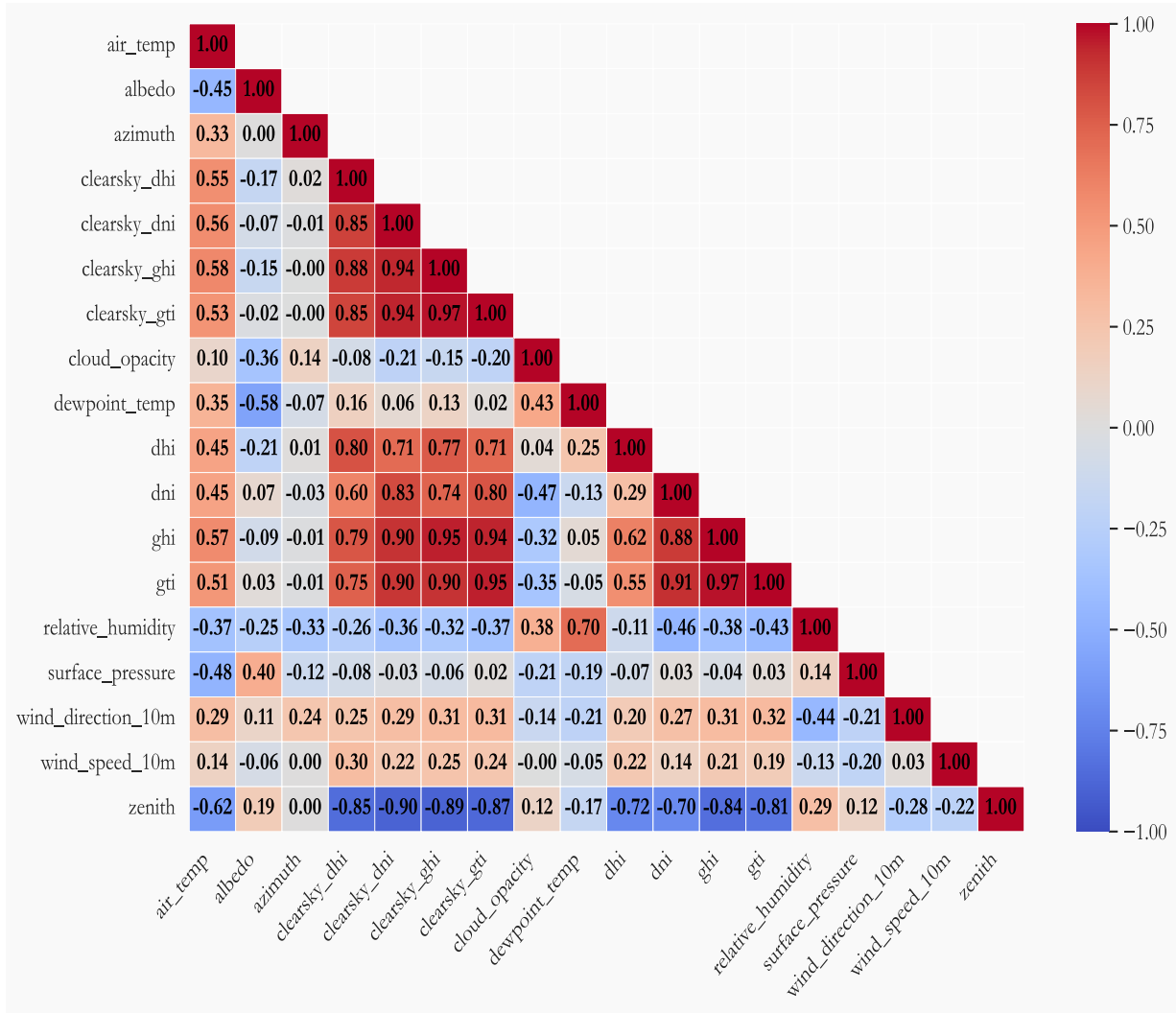


FIGURE 11. Correlation of Solcast weather elements for the rooftop site at VPN Technologies, Pretoria, South Africa.

STC, commonly 1000 W/m^2 . The coefficient k_{RH} quantifies the proportional reduction in power output due to relative humidity effects, and RH is the relative humidity expressed as a fraction or percentage (appropriately normalized). These corrections add physically meaningful environmental dependencies often absent or too simplified in PVLIB.

6) DETAILED POWER SCALING AND LOSSES

The power is scaled by the number of panels, inverter efficiency and a preset 1% loss factor accounting for wiring and other system losses. This attention to system-level losses improves the accuracy of AC power and energy estimation.

7) HIGHER TEMPORAL RESOLUTION AND AGGREGATION

Utilizing five-minute interval data with accurate resampling to hourly and daily aggregates preserves temporal variability and enhances fidelity. This approach outperforms PVLIB

methods that rely on interpolated or smoothed hourly data, which can obscure short-term fluctuations.

8) INTEGRATION OF REAL WEATHER PARAMETERS WITH MODEL PHYSICS

The method incorporates fundamental meteorological parameters—including DNI, DHI, GHI, cloud opacity, ambient temperature, relative humidity and surface albedo—within a transparent and physics-grounded modeling framework. Unlike closed-box machine learning approaches, it maintains interpretability by aligning model components with established principles. Notably, cloud opacity in the OSM-MEPS model serves as a physically justified attenuation factor for direct normal irradiance (DNI), accounting for atmospheric scattering and absorption rather than relying on generic statistical adjustments. In addition, temperature and humidity exert direct effects on the semiconductor behavior of photovoltaic (PV) cells by modulating carrier mobility

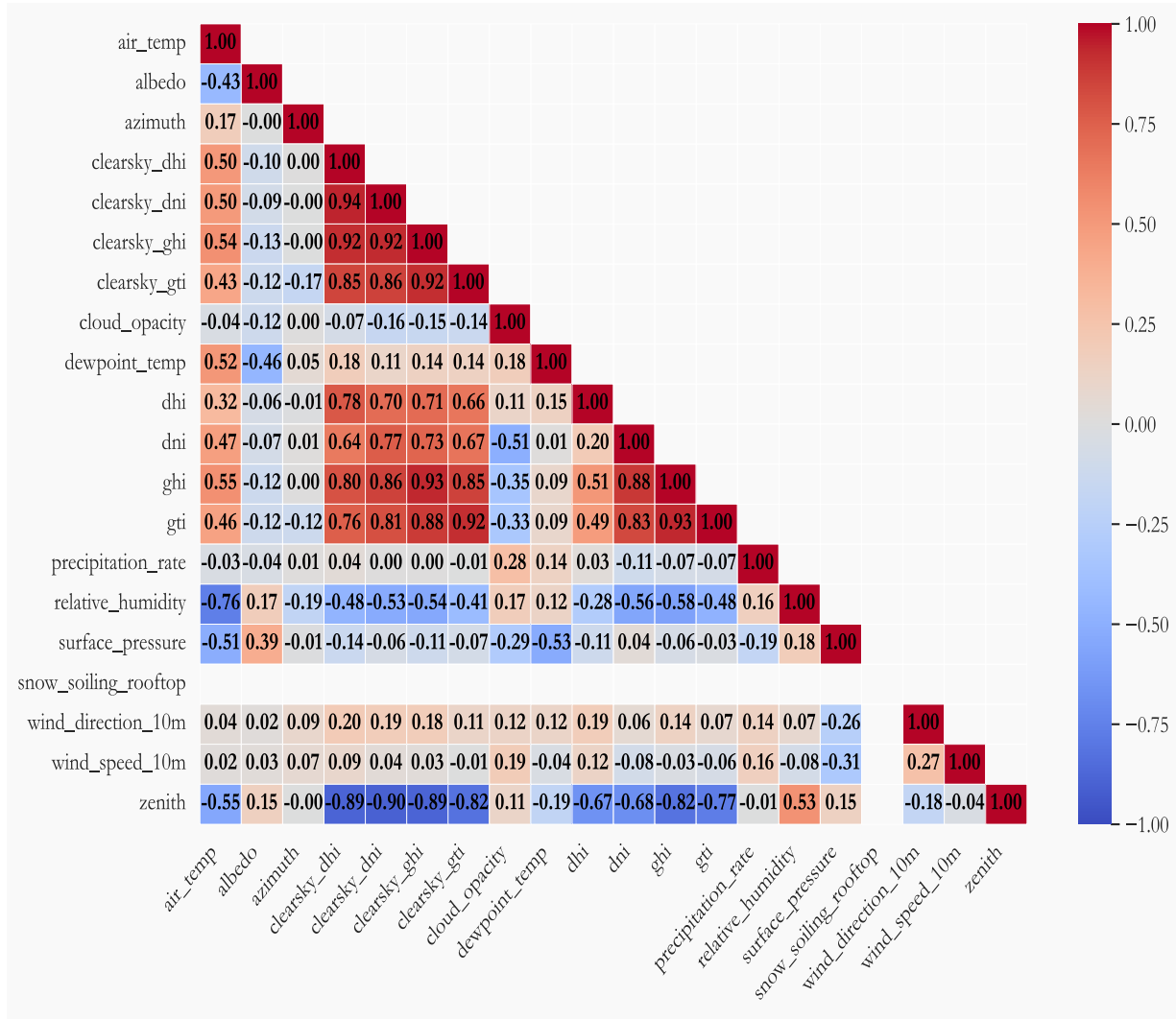


FIGURE 12. Correlation of Solcast weather elements for the rooftop site at 367 Glen Osmond, Adelaide, Australia.

and recombination rates, thereby influencing conversion efficiency and electrical output.

D. OSM-MEPS BASIC MODEL: SERRES-C SOLAR PV POWER PLANT

1) PANEL SPECIFICATIONS

$$P_{\text{panel}}^{\text{max}} = 390 \text{ W} \quad (7)$$

$$A_{\text{panel}} = 1.6 \text{ m}^2 \quad (8)$$

$$\alpha_T = -0.45 \text{ (\%/}^\circ\text{C)} \quad (9)$$

$$T_{\text{NOCT}} = 45 \text{ }^\circ\text{C} \quad (10)$$

$$I_{\text{STC}} = 1000 \text{ W/m}^2 \quad (11)$$

2) PV SYSTEM SPECIFICATIONS

$$P_{\text{sys}} = 1,010,880 \text{ W} \quad (12)$$

$$N_{\text{panels}} = \frac{P_{\text{sys}}}{P_{\text{panel}}^{\text{max}}} = \frac{1,010,880}{390} \approx 2,592 \quad (13)$$

3) PANEL ORIENTATION AND INVERTER EFFICIENCY

$$\theta_{\text{tilt}} = 25^\circ \quad (14)$$

$$\theta_{\text{panel azimuth}} = 180^\circ \text{ (south-facing)} \quad (15)$$

$$\eta_{\text{inverter}} = 0.94 \quad (16)$$

4) SOLAR GEOMETRY MODELING

The angle of incidence (AOI) is given by [71]:

$$\begin{aligned} \theta_{\text{AOI}} = \cos^{-1} \big(& \cos(\theta_{\text{zenith}}) \cos(\theta_{\text{tilt}}) \\ & + \sin(\theta_{\text{zenith}}) \sin(\theta_{\text{tilt}}) \\ & + \cos(\theta_{\text{sun azimuth}} - \theta_{\text{panel azimuth}}) \big) \end{aligned} \quad (17)$$

Values are clipped between 0° and 90°

5) IRRADIANCE CALCULATIONS ON THE TILTED SURFACE [71]

$$I_{\text{poa, direct}} = I_{\text{DNI}} \cos(\theta_{\text{AOI}}) \quad (18)$$

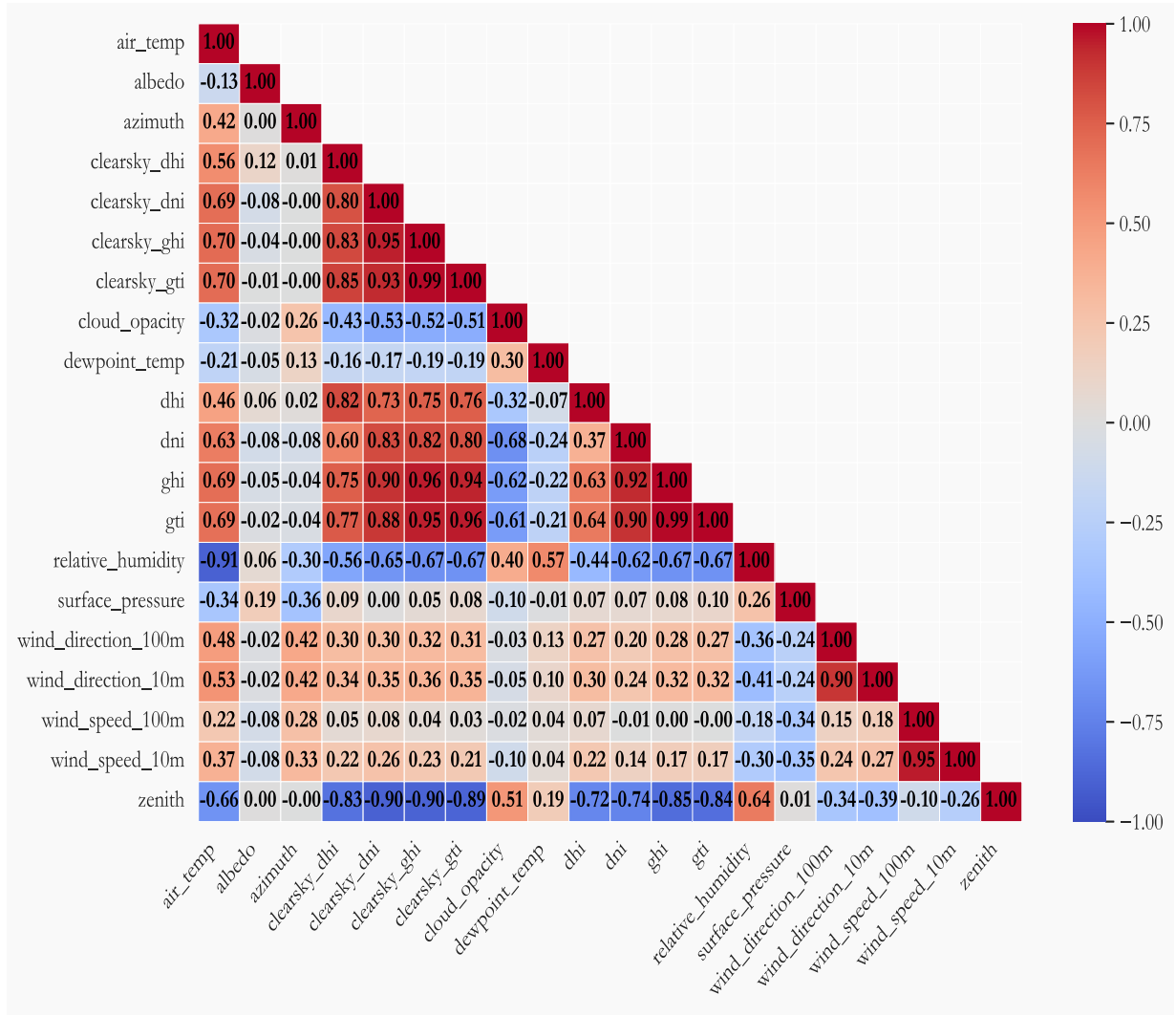


FIGURE 13. Correlation of Solcast weather elements for Victoria Views, Kisumu, Kenya.

$$I_{\text{poa, sky diffuse}} = I_{\text{DHI}} \frac{(1 + \cos(\theta_{\text{tilt}}))}{2} \quad (19)$$

$$I_{\text{poa, diffuse}} = I_{\text{GHI}} \beta \frac{(1 - \cos(\theta_{\text{tilt}}))}{2} \quad (20)$$

$$I_{\text{poa, total}} = I_{\text{poa, direct}} + I_{\text{poa, sky diffuse}} + I_{\text{poa, diffuse}} \quad (21)$$

6) MODULE TEMPERATURE [71], [73], [74]

$$T_{\text{module}} = T_{\text{NOCT}} + \frac{I_{\text{poa, total}}}{1000} (28 - T_{\text{air}}) \quad (22)$$

7) DC POWER OUTPUT CALCULATION [71], [73], [74]

$$P_{\text{panel}} = P_{\text{panel}}^{\text{max}} (1 + \alpha_T (T_{\text{module}} - T_{\text{NOCT}})) \quad (23)$$

$$P_{\text{DC}} = P_{\text{panel}} \frac{I_{\text{poa, total}}}{I_{\text{STC}}} \quad (24)$$

8) AC POWER OUTPUT VIA INVERTER [71]

$$P_{\text{AC}} = P_{\text{DC}} \eta_{\text{inverter}} \quad (25)$$

$$P_{\text{scaled}} = P_{\text{AC}} \times N_{\text{panels}} \quad (26)$$

9) ENERGY PRODUCTION CALCULATION

$$E_{\text{daily}} = \sum P_{\text{scaled}} \Delta t \quad (27)$$

$$kWh_{\text{total}} = \sum E_{\text{daily}} / 1000 \quad (28)$$

where Δt is the 5-minute time step converted to hours.

E. RESIDENTIAL ROOF-TOP SOLAR PV ENERGY MODEL FOR NJAGU-RUIRU ESTATE, KENYA

This section presents the physics-based and mathematical modeling of a photovoltaic (PV) system, incorporating key parameters such as panel area, temperature effects, solar irradiance and inverter efficiency. The model calculates solar PV energy output by accounting for changes in panel orientation, tilt and weather conditions such as irradiance and temperature. For this study, 40 m² of the available 56.04 m² rooftop area—identified via Google Earth (Figure 24)—is allocated for PV installation. The flat roof allows height

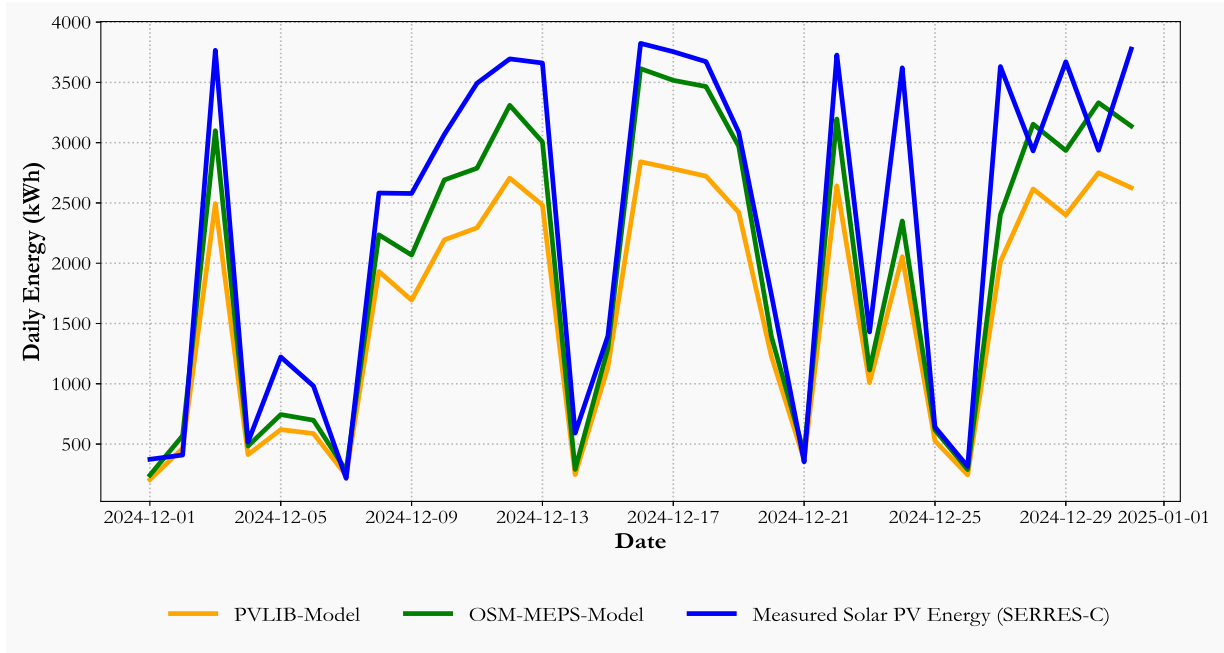


FIGURE 14. Comparison of Serres-C measured energy generation [76] with PVLIB and OSM-MEPS model results for December 2024.

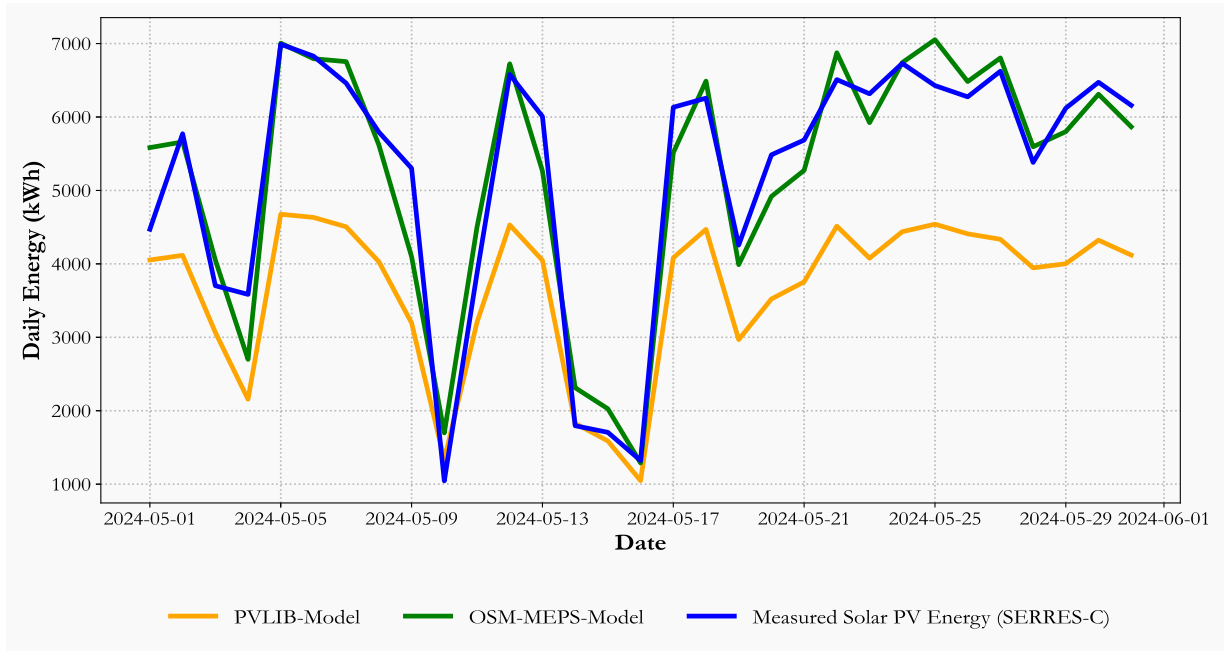


FIGURE 15. Comparison of Serres-C measured energy generation [76] with PVLIB and OSM-MEPS model results for May 2024.

adjustments to mitigate shading. A tilt angle of 22.5° is selected for feasibility analysis. Using the OSM-MEPS framework, scenario iterations are performed to identify optimal panel orientations, with insights supporting future rooftop solar PV energy planning.

1) PANEL SPECIFICATIONS

The following parameters define the PV panel characteristics:

$$P_{\text{panel}}^{\text{max}} = 350 \text{ W} \quad (29)$$

$$A_{\text{panel}} = 1.6 \text{ m}^2 \quad (30)$$

$$\alpha_T = -0.45 (\%/^\circ\text{C}) \quad (31)$$

$$T_{\text{NOCT}} = 45^\circ\text{C} \quad (32)$$

$$I_{\text{STC}} = 1000 \text{ W/m}^2 \quad (33)$$

2) PV SYSTEM SPECIFICATIONS

The overall system specifications are as follows:

$$A_{\text{total}} = 40 \text{ m}^2 \quad (34)$$

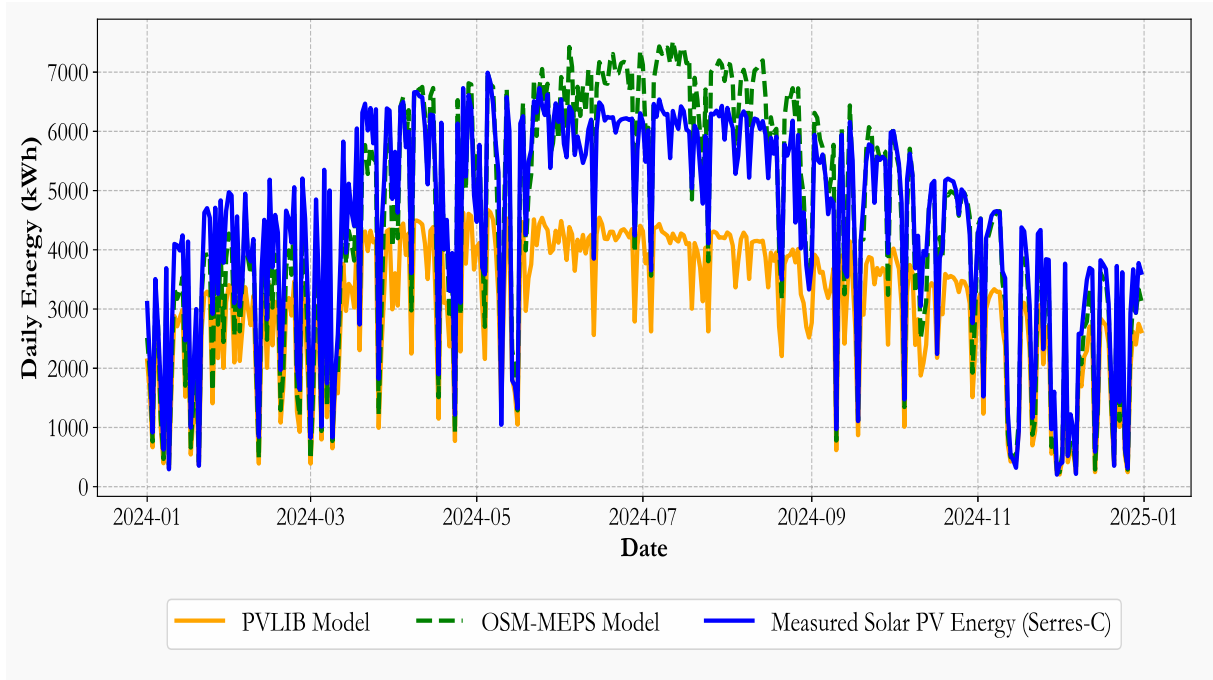


FIGURE 16. Comparison of PVLIB and OSM-MEPS model solar PV energy results with Serres-C PV power plant output, Serres, Greece.

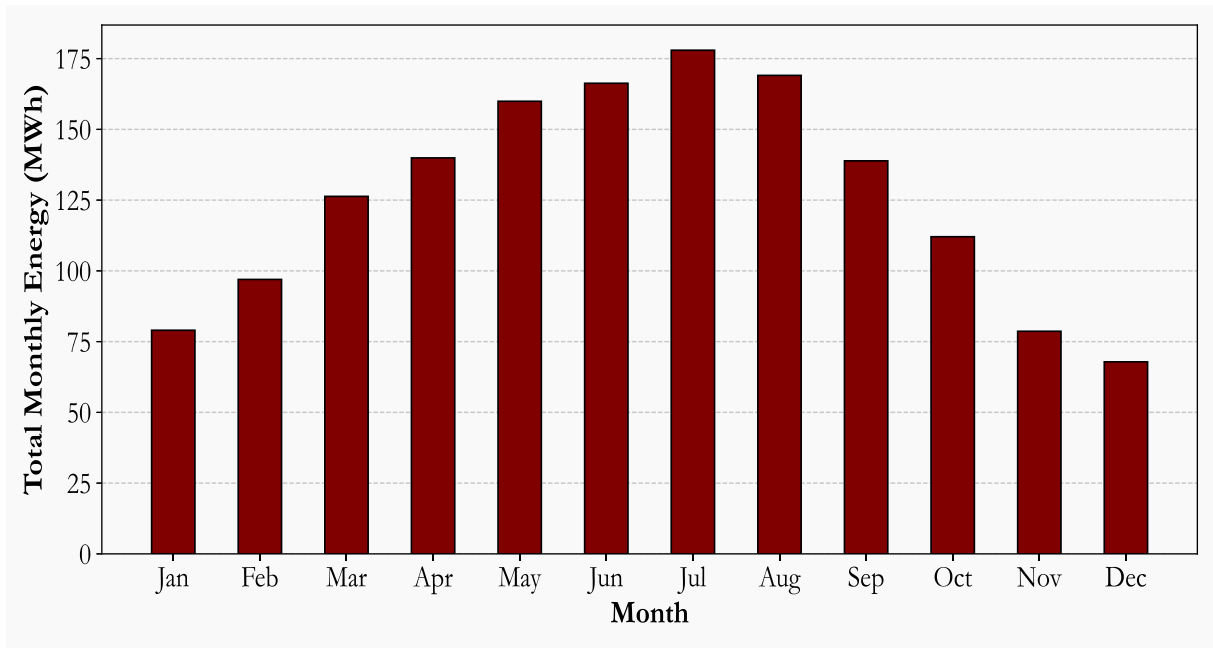


FIGURE 17. Monthly solar energy for Serres-C, Greece, based on GSA data [37].

$$N_{\text{panels}} = \frac{A_{\text{total}}}{A_{\text{panel}}} = \frac{40}{1.6} = 25 \quad (35)$$

$$\eta_{\text{inverter}} = 0.94 \quad (38)$$

3) PANEL ORIENTATION AND INVERTER EFFICIENCY [75]

The PV system is oriented with the following parameters, and the inverter has an efficiency of:

$$\theta_{\text{tilt}} = 22.5^\circ \quad (36)$$

$$\theta_{\text{panel azimuth}} = 90^\circ \quad (37)$$

4) SOLAR GEOMETRY MODELING

The angle of incidence (AOI) is calculated as [71]:

$$\begin{aligned} \theta_{\text{AOI}} = & \cos^{-1} (\cos(\theta_{\text{zenith}}) \cos(\theta_{\text{tilt}}) \\ & + \sin(\theta_{\text{zenith}}) \sin(\theta_{\text{tilt}}) \cos(\theta_{\text{sun azimuth}} - \theta_{\text{panel azimuth}})) \end{aligned} \quad (39)$$

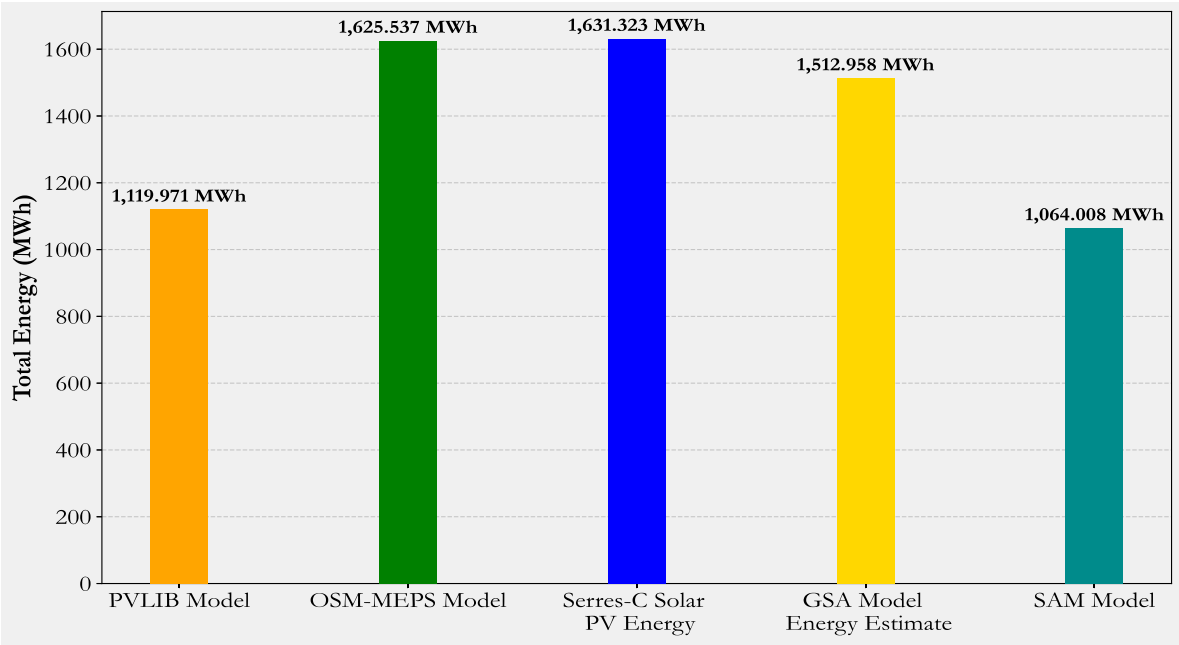


FIGURE 18. Comparison of modeled 2024 solar PV energy yield—using PVLIB, OSM-MEPS, GSA and SAM—with actual measured energy output (MWh) at the Serres-C power plant.

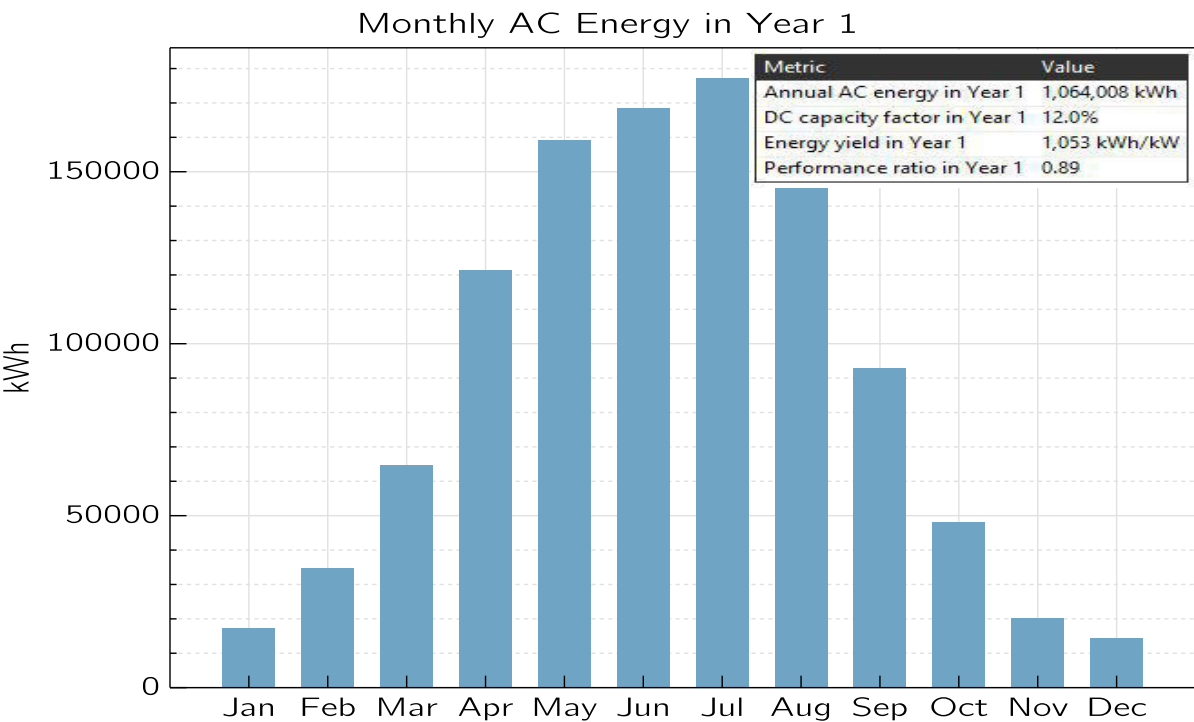


FIGURE 19. Monthly energy estimate for the Serres-C site using NSRDB Typical Meteorological Year (TMY) data in SAM.

AOI values are clipped between 0° and 90° to maintain physical validity in irradiance calculations, especially for mono-facial solar PV panels. An AOI of 0° means sunlight strikes the panel surface perpendicularly, maximizing energy absorption. On the contrary, an AOI of 90° means the sun’s

rays are parallel to the panel surface, resulting in minimal or no energy capture since the effective irradiance approaches zero. Clipping AOI values ensures that the model reflects realistic solar incidence conditions. However, this clipping does not exclude albedo (reflected) or diffuse radiation

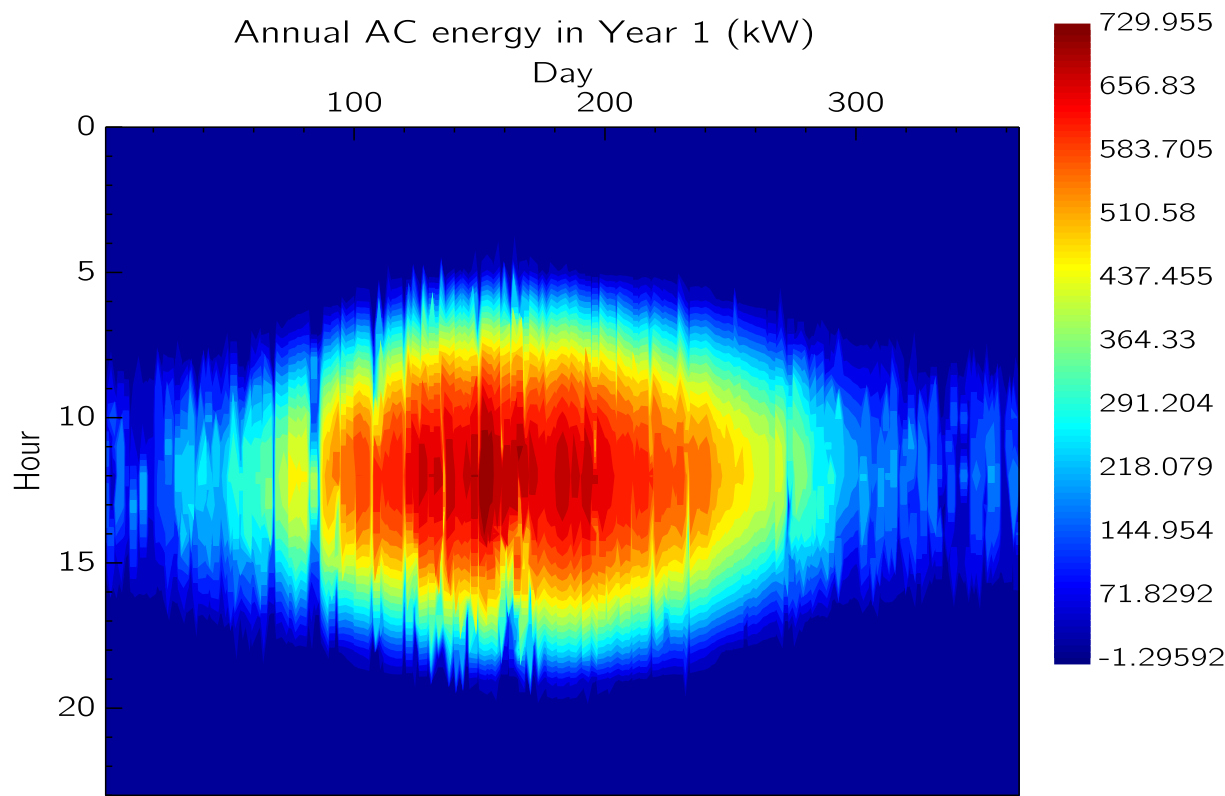


FIGURE 20. Hourly energy heatmap for the Serres-C site using NSRDB Typical Meteorological Year (TMY) data in SAM.

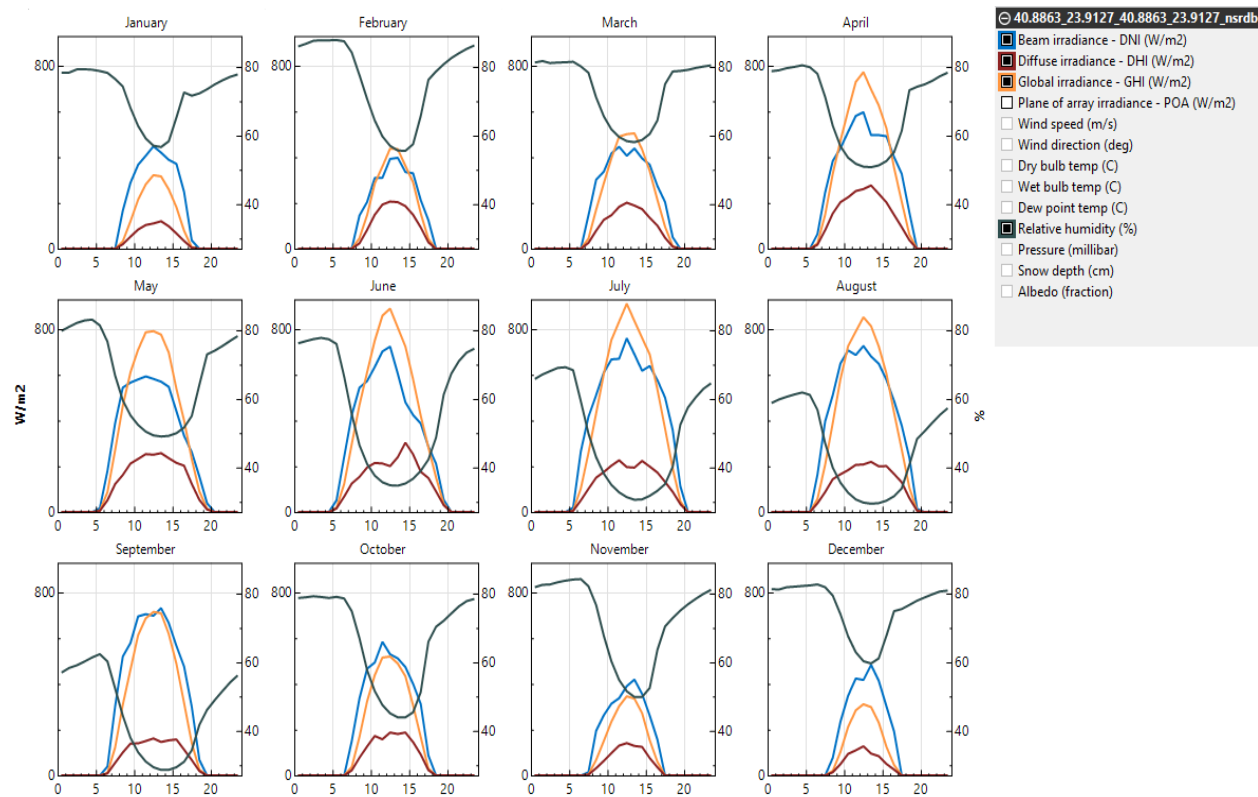


FIGURE 21. Relationship between irradiance and relative humidity in SAM for the Serres-C site using an NSRDB TMY dataset.

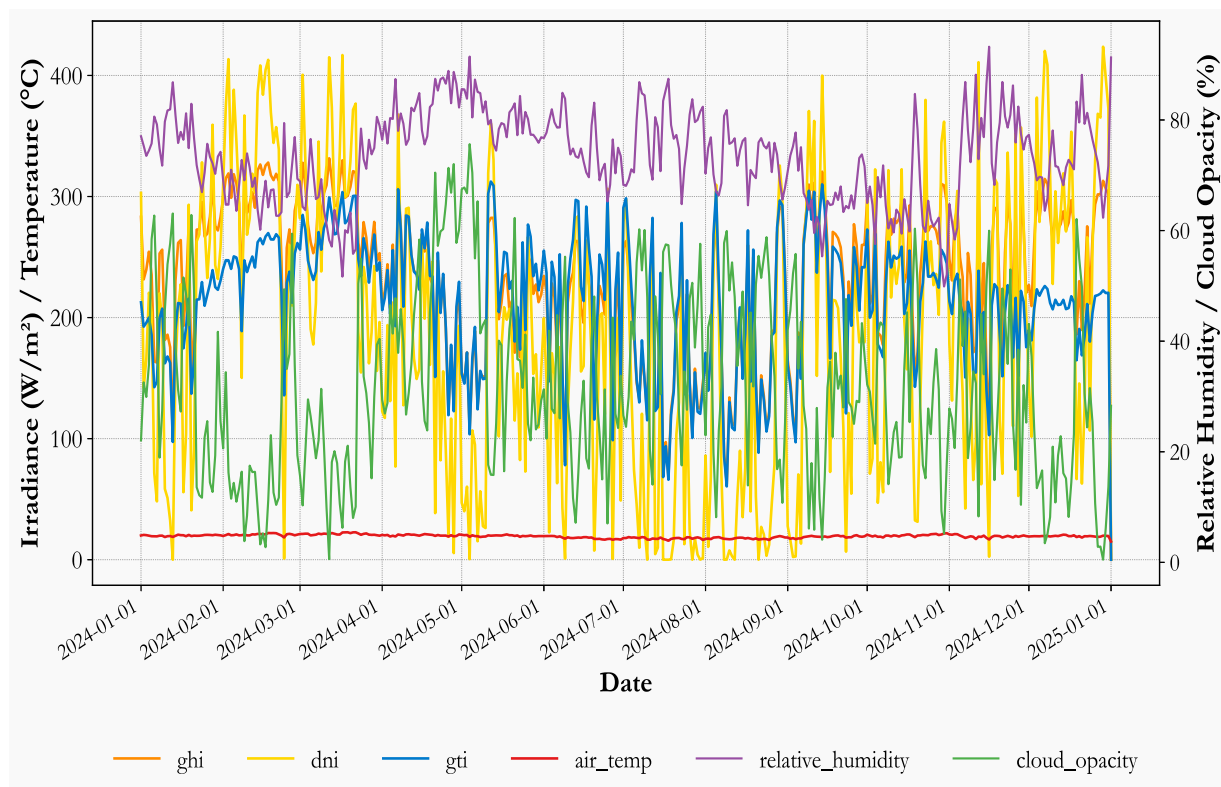


FIGURE 22. 24-hour averaged Solcast key weather parameters for 2024 at Njagu-Mhasibu-Ruiru Estate, Kenya.

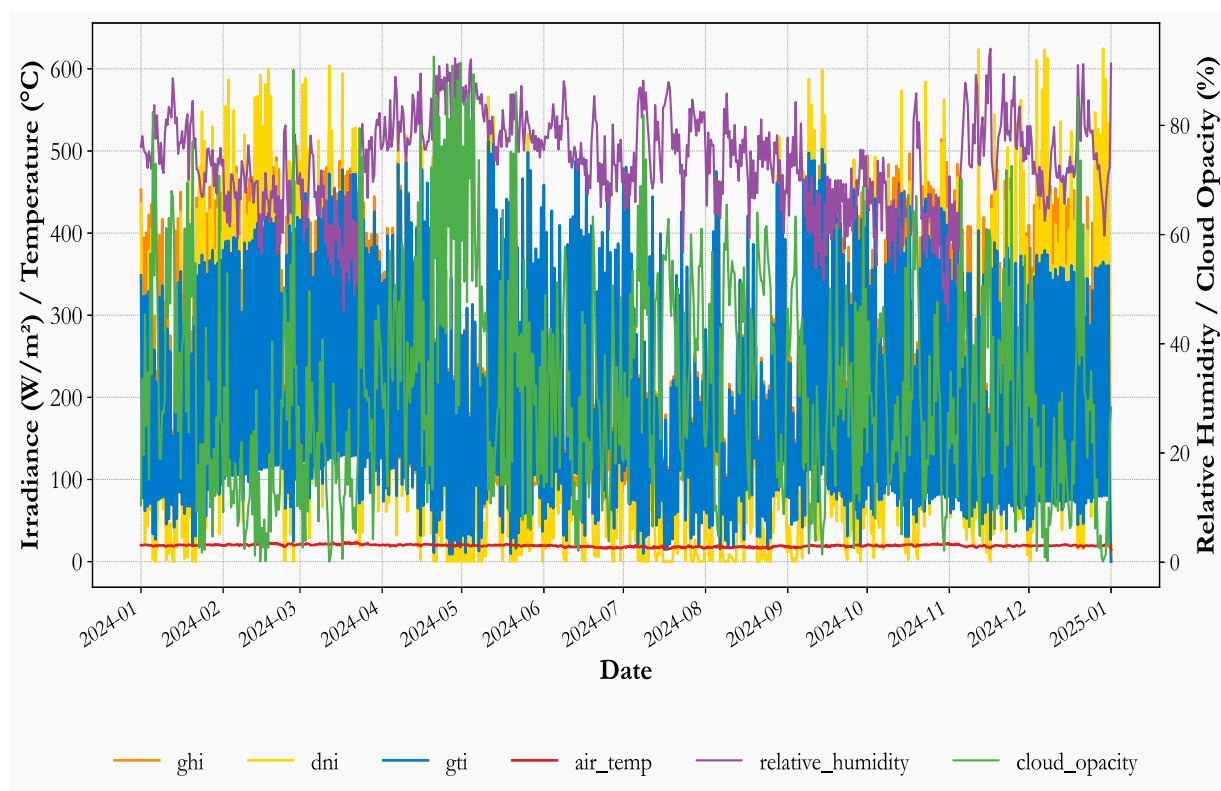


FIGURE 23. 12-hour averaged Solcast key weather parameters for 2024 at Njagu-Mhasibu-Ruiru Estate, Kenya.



FIGURE 24. Site location of the proposed rooftop solar PV installation site and Serres-C solar PV plant © Google Earth.

effects, as these components are treated separately within the solar PV irradiance modeling.

5) IRRADIANCE CALCULATIONS ON THE TILTED SURFACE [71]

The irradiance components on the tilted surface are:

$$I_{\text{poa, direct}} = I_{\text{DNI}} \cos(\theta_{\text{AOI}}) \quad (40)$$

$$I_{\text{poa, sky diffuse}} = I_{\text{DHI}} \frac{(1 + \cos(\theta_{\text{tilt}}))}{2} \quad (41)$$

$$I_{\text{poa, diffuse}} = I_{\text{GHI}} \beta \frac{(1 - \cos(\theta_{\text{tilt}}))}{2} \quad (42)$$

$$I_{\text{poa, total}} = I_{\text{poa, direct}} + I_{\text{poa, sky diffuse}} + I_{\text{poa, diffuse}} \quad (43)$$

6) MODULE TEMPERATURE

The module temperature is estimated using [71], [73], [74]:

$$T_{\text{module}} = T_{\text{NOCT}} + \frac{I_{\text{poa, total}}}{1000} (28 - T_{\text{air}}) \quad (44)$$

7) DC POWER OUTPUT CALCULATION [71], [73], [74]

The DC power output is calculated as follows:

$$P_{\text{panel}} = P_{\text{panel}}^{\text{max}} (1 + \alpha_T (T_{\text{module}} - T_{\text{NOCT}})) \quad (45)$$

$$P_{\text{DC}} = P_{\text{panel}} \frac{I_{\text{poa, total}}}{I_{\text{STC}}} \quad (46)$$

8) AC POWER OUTPUT VIA INVERTER

The AC power output is determined by [75]:

$$P_{\text{AC}} = P_{\text{DC}} \eta_{\text{inverter}} \quad (47)$$

$$P_{\text{scaled}} = P_{\text{AC}} \times N_{\text{panels}} \quad (48)$$

9) ENERGY PRODUCTION CALCULATION

Finally, the daily and total energy production are:

$$E_{\text{daily}} = \sum P_{\text{scaled}} \Delta t \quad (49)$$

$$kWh_{\text{total}} = \sum E_{\text{daily}} / 1000 \quad (50)$$

where Δt is the 5-minute time step converted to hours.

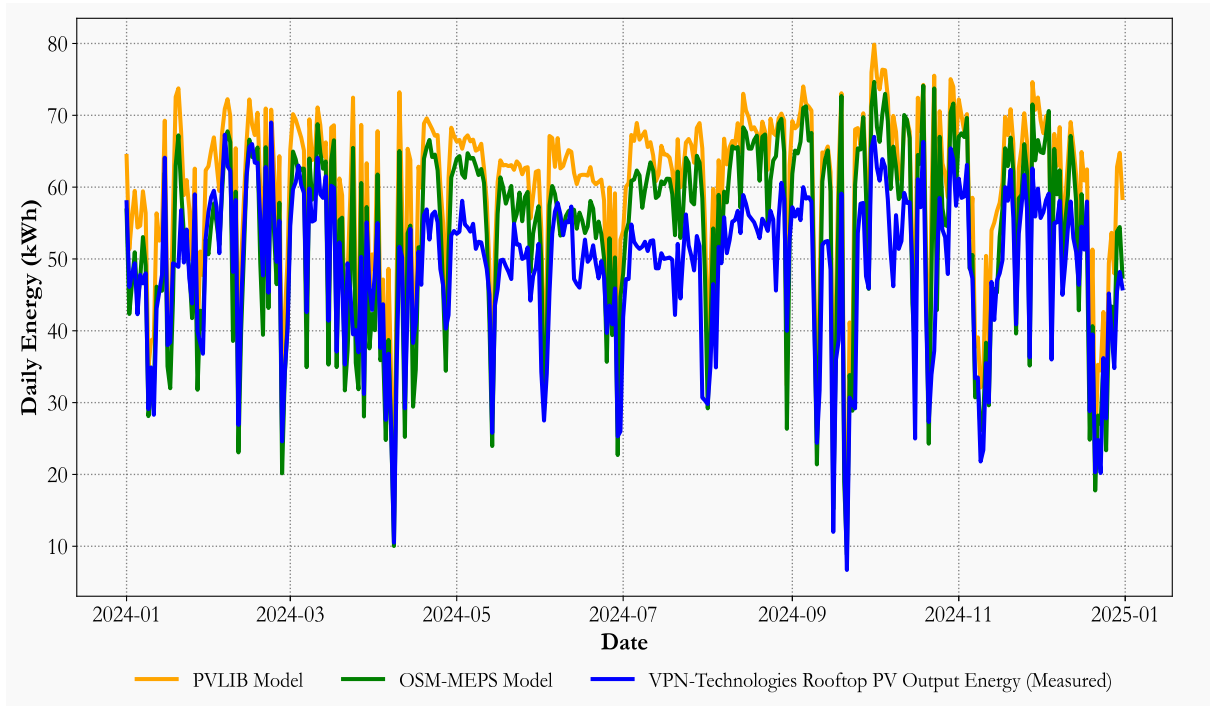


FIGURE 25. Comparison of PVLIB and OSM-MEPS model solar PV energy results with VPN Technologies rooftop PV energy data, Pretoria, South Africa.

A_{total} represents the total available area. A_{panel} denotes the area of each individual panel. The number of panels in the system is denoted as N_{panels} . The angle of incidence is represented by θ_{AOI} . Irradiance values are characterized by I_{DNI} for direct normal irradiance, I_{DHI} for diffuse horizontal irradiance, and I_{GHI} for global horizontal irradiance. The albedo is indicated by β . Air temperature is represented by T_{air} . The irradiance at standard testing conditions I_{STC} is normally 1000 W/m^2 at 25°C cell temperature and air mass of 1.5 (AM1.5). The inverter efficiency is indicated by η_{inverter} . Finally, Δt represents the time steps of power computation. Model implementation was done in Python. The tilt-azimuth optimal orientation was computed step by step as seen in algorithm 1.

V. RESULTS, ANALYSIS AND DISCUSSION

A. OSM-MEPS MODEL VALIDATION

An annual energy output graph for the Serres-C site in Greece was generated to evaluate the OSM-MEPS model's ability to capture seasonal production variations (Figure 16). This output was benchmarked against results from GSA, PVLIB and SAM models (Figures 17, 18, 19 and 20). To assess the impact of atmospheric conditions, we analyzed the relationship between relative humidity and irradiance, deriving an attenuation factor model based on relative humidity (Equation 6, Figure 21). Validation was extended to sites in South Africa, Australia and Kenya, covering

both hemispheres with diverse weather, geography and solar profiles to test model robustness.

The results for Serres-C, Greece, are presented in Figure 16, while Figure 25 shows the energy output for a rooftop site in Pretoria, South Africa. The rooftop installation at 367 Glen Osmond is depicted in Figure 26, with its corresponding energy output illustrated in Figure 27. Figure 28 displays the energy output of a medium-shaded PV system located in Kisumu, Kenya, near the equator. Error metrics used to assess OSM-MEPS model accuracy against the actual solar PV output energy, for each site, are discussed in Section VI. The energy production profile in the Northern Hemisphere generally exhibits a bell-shaped curve, as seen in Figure 29, whereas in the Southern Hemisphere, the trend resembles an inverted bell curve, as shown in Figure 30.

B. DAILY SOLAR PV ENERGY PRODUCTION FOR VARIOUS TILT-AZIMUTH ANGLES

When a solar panel (or POA) is positioned straight up (at a 90° tilt), it mostly catches scattered sunlight instead of direct rays from the sun. Because of this, it produces less energy throughout the year in both the northern and southern hemispheres. The use of vertical panels includes: building facades, noise barriers and urban settings where roof space is limited, enabling efficient space utilization while integrating into existing structures despite lower efficiency. This solar PV energy phenomenon is confirmed



FIGURE 26. 367 Glen Osmond vicinity and rooftop solar PV installation site, Adelaide, Australia (© Google Earth).

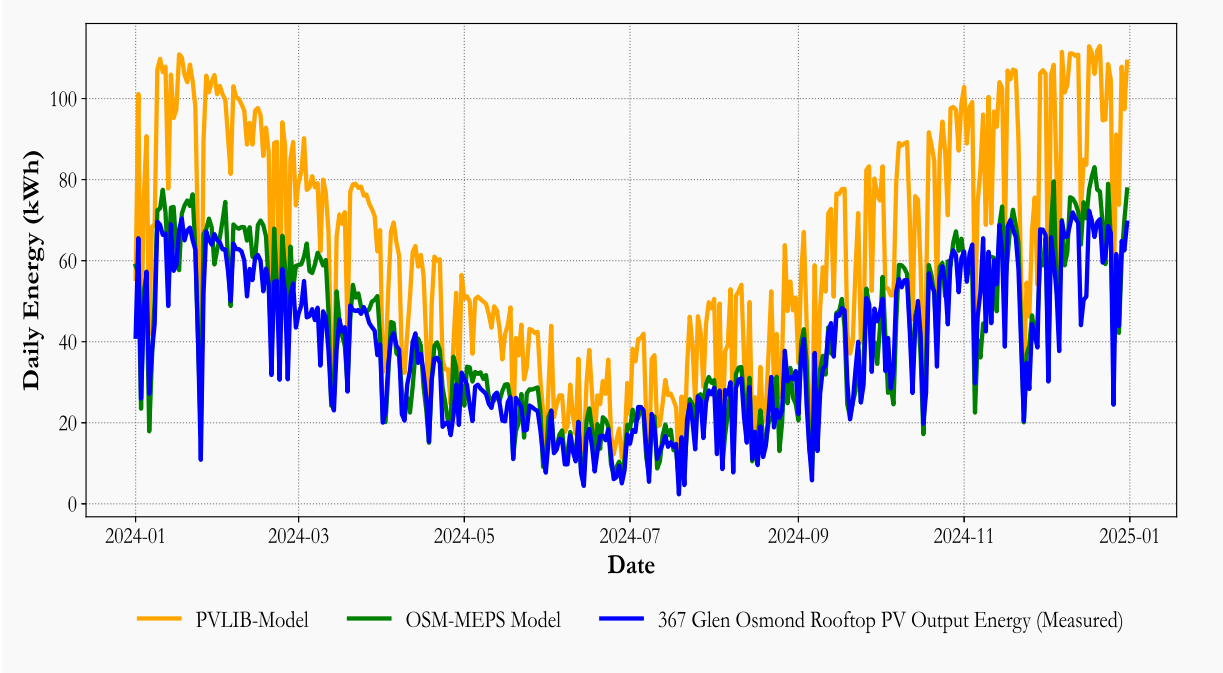


FIGURE 27. Comparison of PVLIB and OSM-MEPS modeled solar PV energy with measured data at 367 Glen Osmond rooftop, Adelaide, Australia.

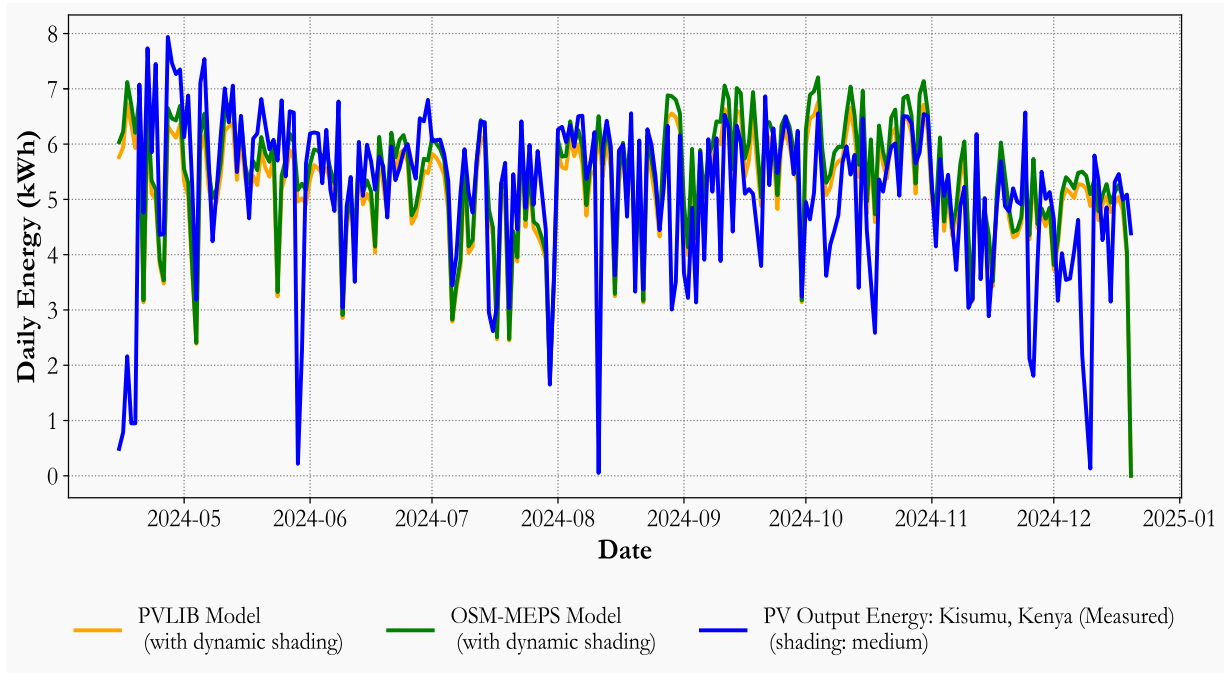


FIGURE 28. Comparison of PVLIB and OSM-MEPS modeled solar PV energy with measured data at Victoria-Views site, Kisumu, Kenya.

by system-scale and rooftop solar PV tilt-azimuth scenarios visualized in Figures 29 and 30.

In contrast, tilt angles between 0° and 45° , as illustrated in Figure 29, demonstrate a clear seasonal pattern, with enhanced energy generation during successive months—April to October 2024 in the Northern Hemisphere, and January to March and September to December 2024 in the Southern Hemisphere—as shown in Figure 30. A solar panel tilted at 180° (facing downward) receives minimal direct sunlight but can still generate some energy by capturing diffuse sky radiation and reflected ground radiation (albedo). This limited output is demonstrated in Figure 34.

C. IMPACT OF SOLAR PV ORIENTATION (TILT = 22.5° , AZIMUTH = 90°) AT NJAGU-MHASIBU-RUIRU ESTATE, KENYA

With a 90° azimuth (east-facing), the panels primarily capture morning sunlight but miss the stronger afternoon irradiance in the months where the sun elevation is higher in the sky. This effect is more pronounced during the winter season, as illustrated in Figures 32 and 33, when the sun's elevation angle is lower, leading to reduced effective irradiance. Additionally, equatorial regions often experience increased cloud cover during July and August, further limiting solar energy availability.

D. DAILY SOLAR PV ENERGY PRODUCTION FOR VARIOUS AZIMUTH ANGLES AT A FIXED TILT ANGLE OF 22.5°

A multi-scenario simulation was performed to assess the impact of photovoltaic (PV) array azimuth orientation on energy output, using a fixed tilt angle of 22.5°

across all seasons. The resulting energy output curves are closely grouped, suggesting that small azimuth variations have minimal influence on energy production during the months surrounding the equinoxes—specifically March to May and August to October—as illustrated in Figure 35. During these periods, the sun's trajectory remains relatively high and symmetric about the east–west axis, enabling east-facing panels to receive significant morning irradiance.

However, noticeable differences in energy generation occur with changes in azimuth angle during January–February and November–December. During these months, the sun rises further southeast, and its early morning position is more oblique, making panel output more sensitive to small azimuth shifts. Similarly, around the June solstice (e.g., June to July), the sun rises further northeast, reducing the effectiveness of strictly east-facing orientations in capturing early morning irradiance.

Ruiru, Kenya, lies near the equator, where the sun stays high year-round, providing consistent solar exposure. While a 0° tilt is typical, a slight 5° tilt is preferred to allow rainwater runoff and reduce dust buildup, helping maintain long-term panel efficiency. The optimal POA orientation as a function of tilt and azimuth angle is as shown in Figure 36. The optimal azimuth angle of 180° —which corresponds to a south-facing orientation—reflects the unique solar geometry of equatorial regions like Ruiru, Kenya. Near the equator, the sun passes both north and south of the zenith throughout the year, making both north and south-facing panels viable. However, a south-facing orientation can offer slightly better alignment with the sun's trajectory during

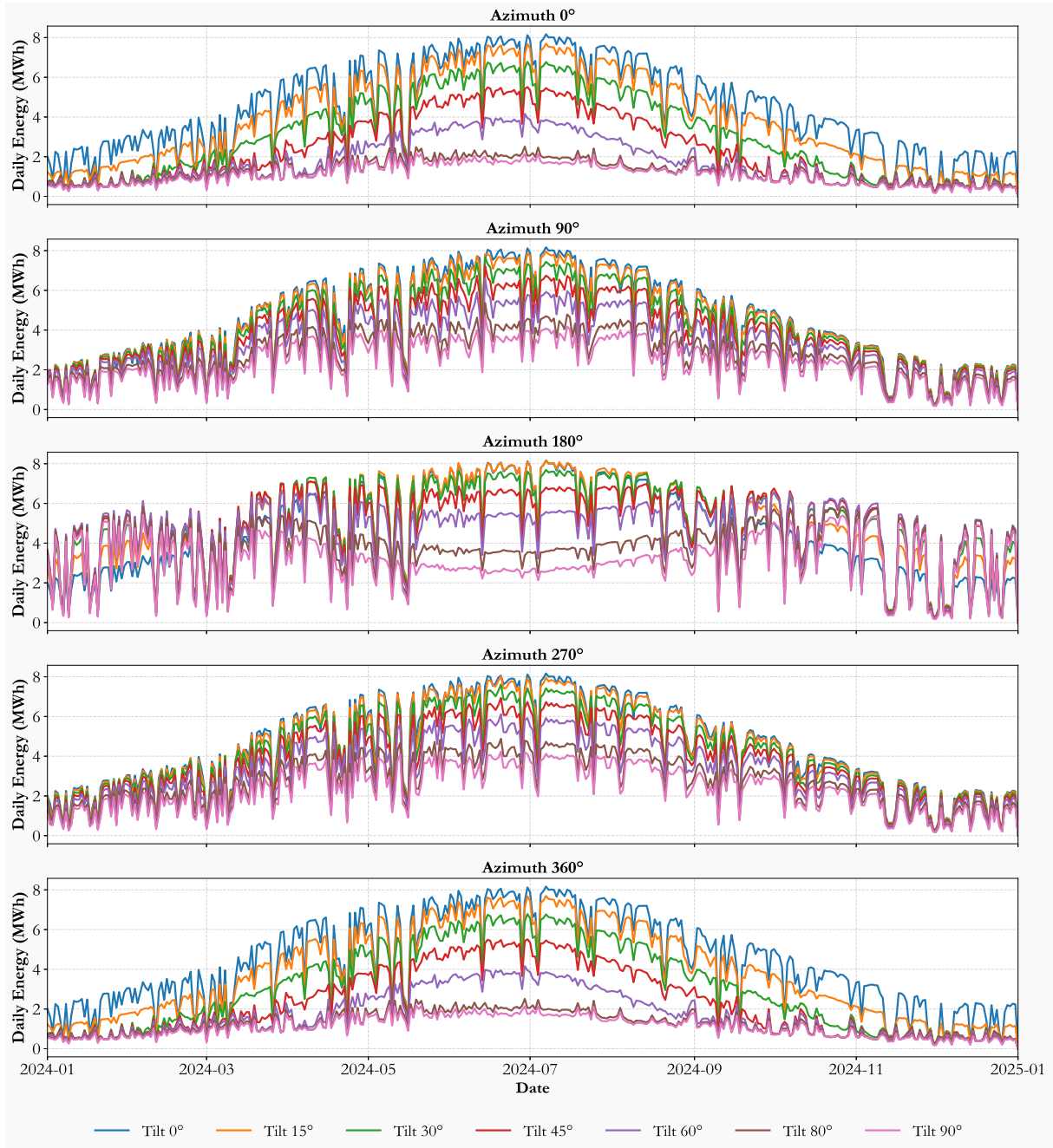


FIGURE 29. Energy production charts for various tilt-azimuth angle combinations of a solar PV array at the Serres-C site, Greece (Northern Hemisphere).

months with higher irradiance, especially from October to February.

The angle selection criteria in Figures 34 and 35 were based on initial scenario simulations that identified configurations with the highest and lowest annual energy outputs. In the final OSM-MEPS iteration, angles near the optimum were included to demonstrate how small orientation changes influence seasonal energy yield and PV performance, supporting scientific interpretation rather than random selection.

E. KEY WEATHER ELEMENTS IN SOLAR PV ENERGY MODELING

The graphs in Figures 22 and 23 depict key weather factors that critically affect solar PV modeling. Higher solar irradiance from September to December drives increased energy output, while lower irradiance from May to August reduces generation. Cloud cover introduces power fluctuations, highlighting the need for energy storage to maintain stability. Although summer offers more sunlight, elevated temperatures decrease PV efficiency, whereas cooler winters,

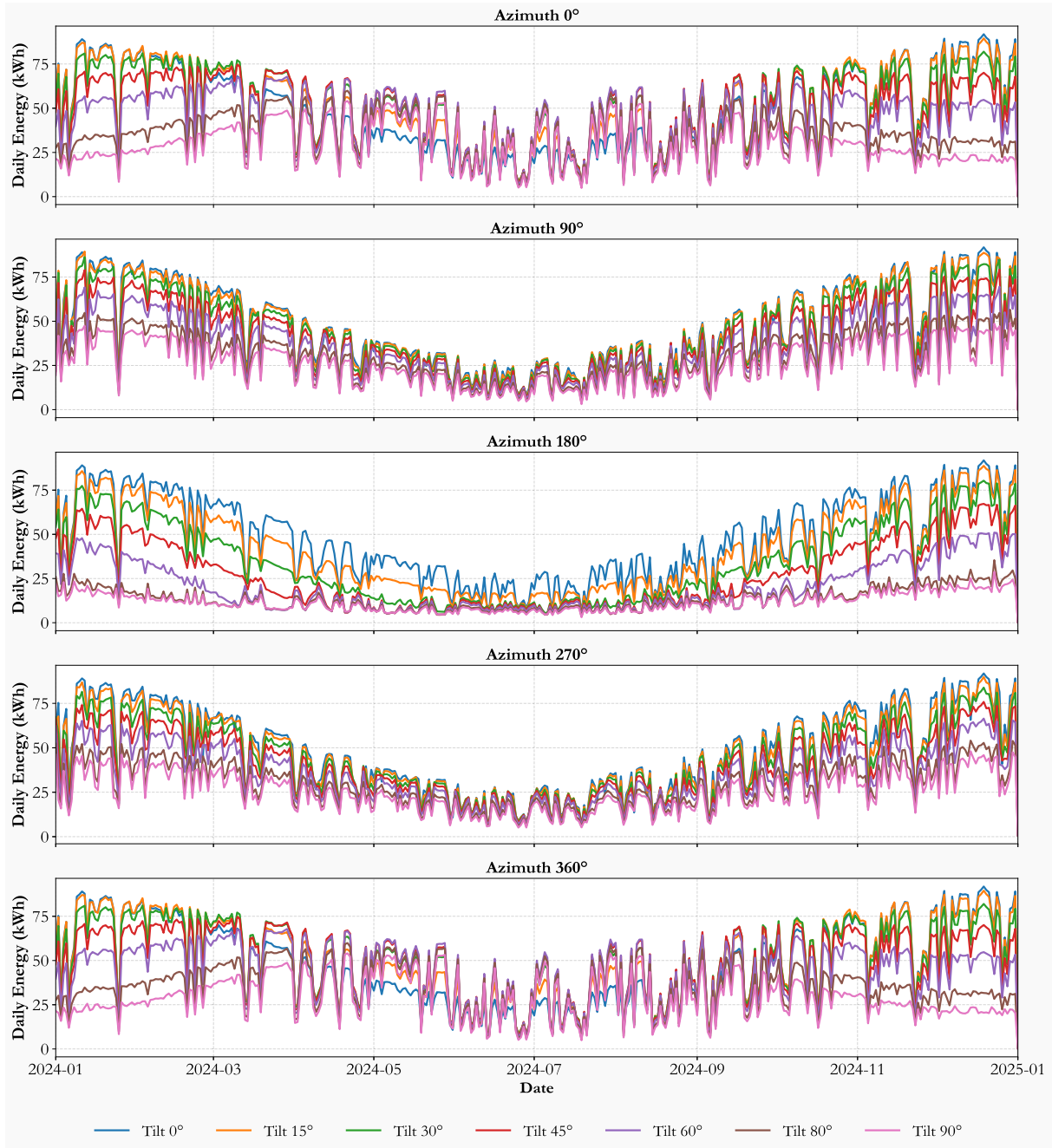


FIGURE 30. Energy production charts for various tilt-azimuth angle combinations of a solar PV array at the Adelaide site, Australia (Southern Hemisphere).

despite less irradiance, improve performance. Understanding this trade-off is essential for accurate forecasting and optimal system design.

Resolution is a crucial aspect of modeling, as it directly impacts the accuracy and detail of solar energy generation analysis. In our study, energy was plotted at hourly intervals by summing 5-minute interval computations, as illustrated in Figure 33. Sampling power at shorter intervals (e.g., every 5 minutes) captures short-term fluctuations in solar

irradiance, cloud cover and shading, providing a more detailed and realistic representation of energy generation. A lower resolution (e.g., daily or monthly averages) would smooth out variations, potentially hiding critical insights such as temporary dips in generation due to passing clouds or efficiency losses during peak temperatures. In contrast, high-resolution data allows for better performance assessment, forecast accuracy and system optimization. For visual presentation and analysis, plotting high-resolution data requires

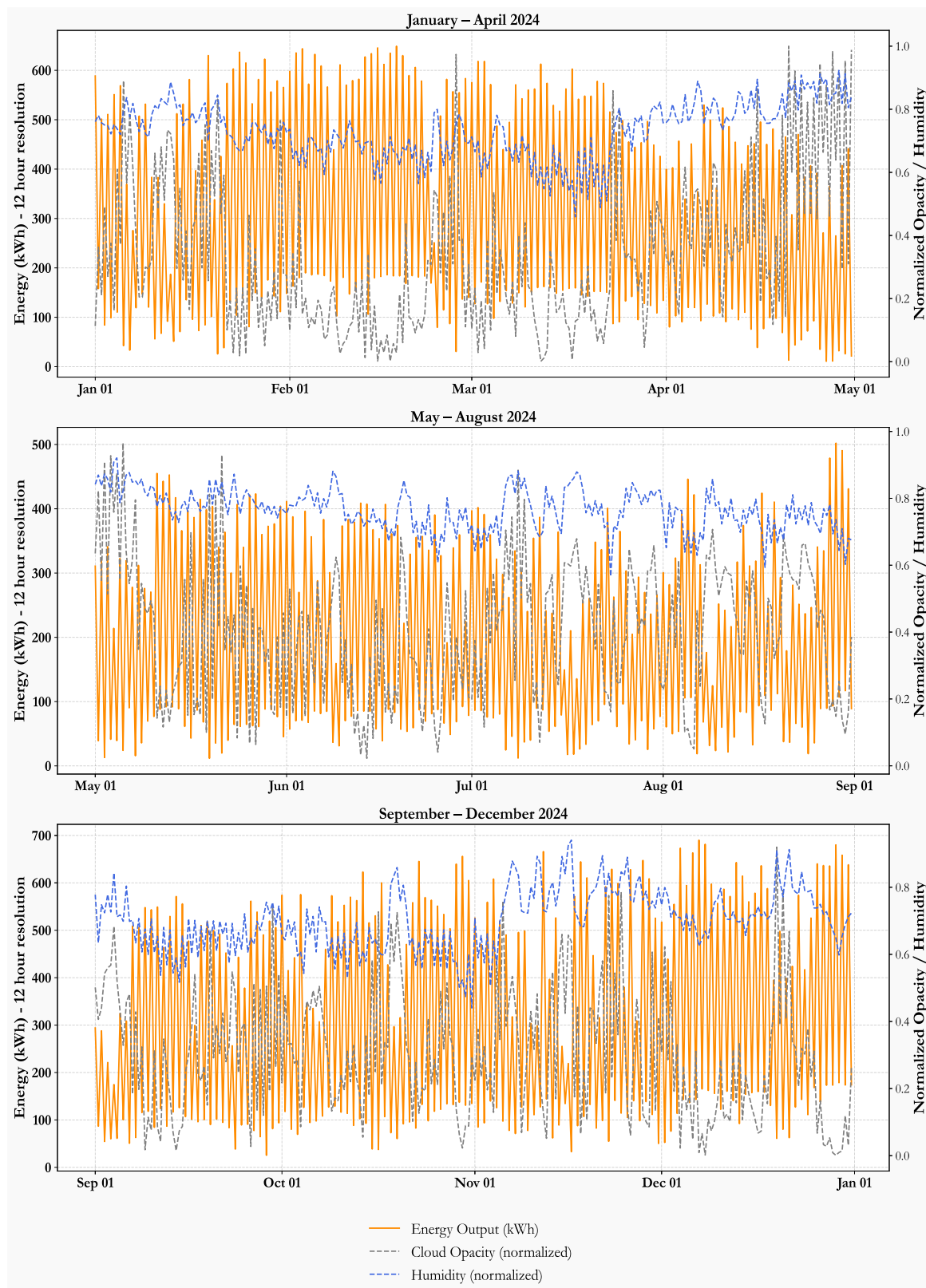


FIGURE 31. Intra-day (12)-hour resolution) energy output for 2024 at Njagu-Mhasibu-Ruiru Estate, Kenya, simulated using the OSM-MEPS model with a panel tilt of 22.5° and azimuth orientation of 184° .

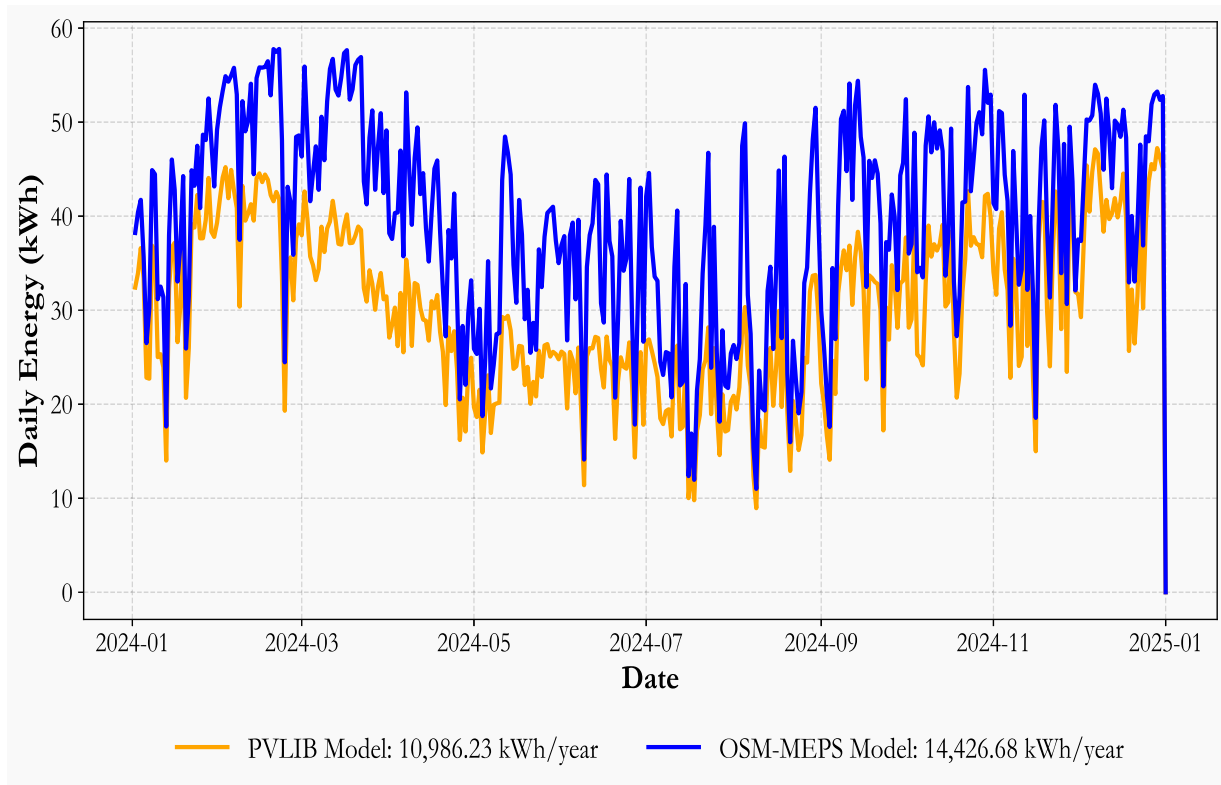


FIGURE 32. Daily energy production for 2024 at Njagu-Mhasibu-Ruiru Estate, Kenya, with a solar panel surface azimuth of 90° and tilt angle of 22.5° .

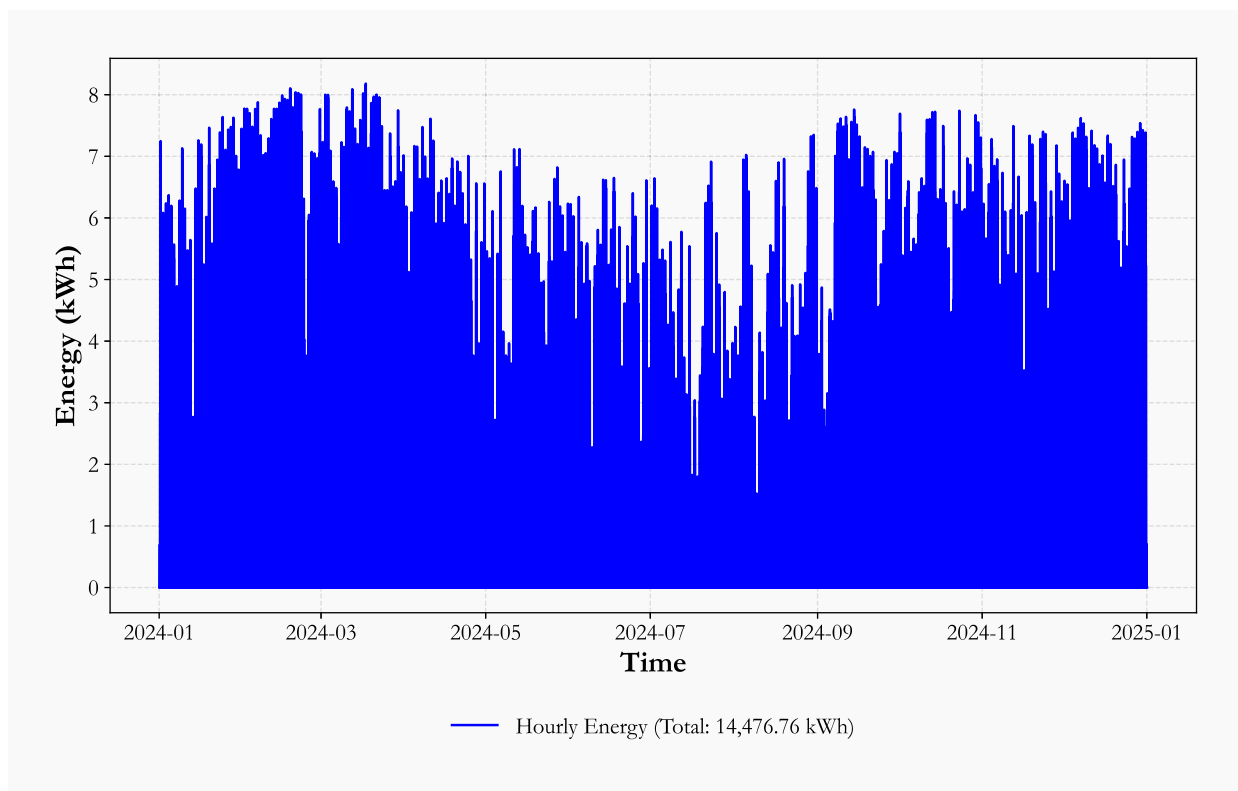


FIGURE 33. Hourly averaged solar energy for 2024, computed at 5-minute resolution using a tilt angle of 22.5° and azimuth of 90° in Njagu-Mhasibu-Ruiru Estate, Kenya.

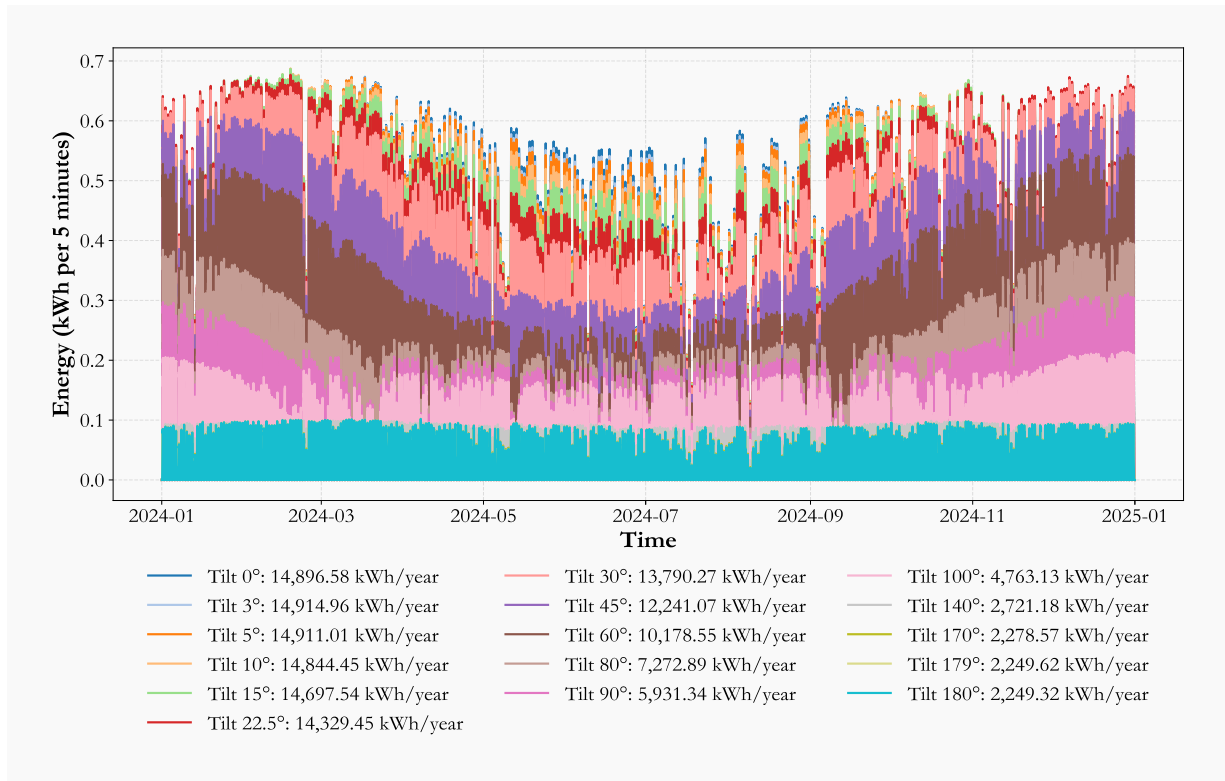


FIGURE 34. Annual 2024 PV energy yield at Njagu-Mhasibu-Ruiru Estate (OSM-MEPS model) for tilt angles 0°–180° at azimuth 180°: optimal tilt (3°) yields 14,914.96 kWh/year; extreme tilt (180°) yields 2,249.32 kWh/year.

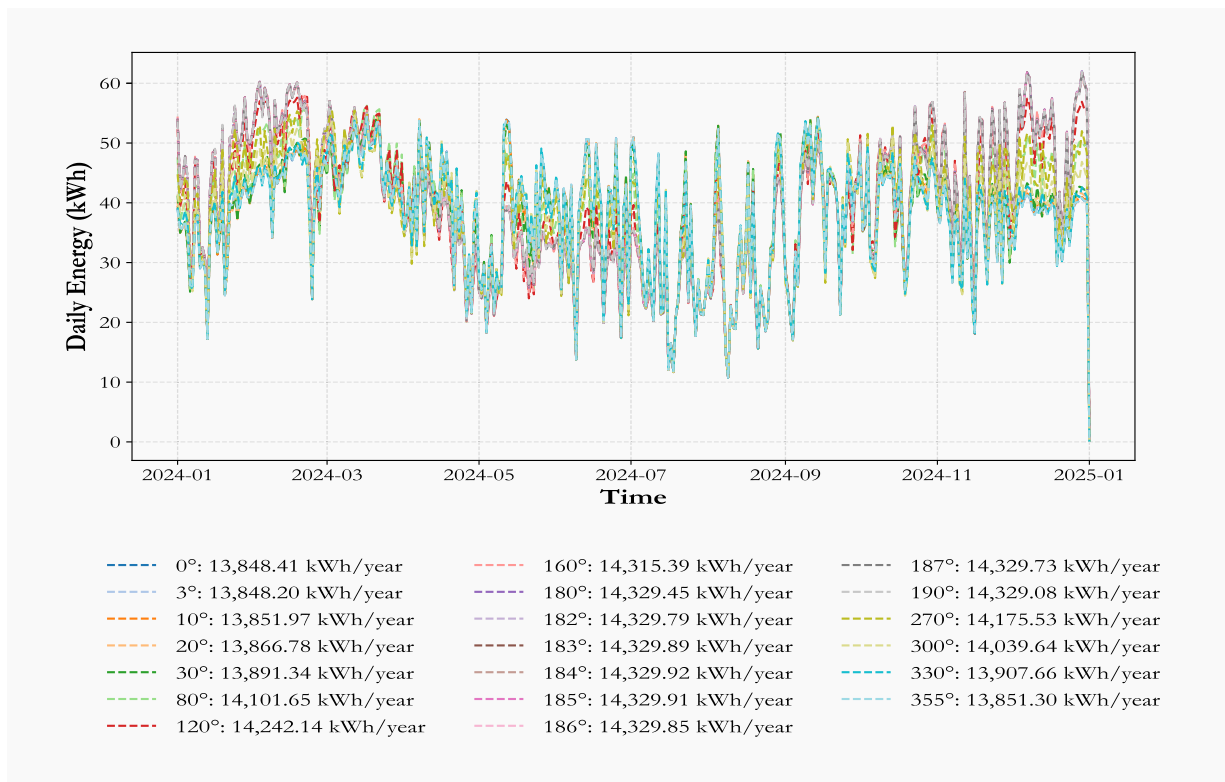


FIGURE 35. Annual PV energy yield in 2024 estimated using the OSM-MEPS model across azimuth angles from 0° to 355° at a fixed tilt of 22.5°. The optimal configuration is 184°, yielding 14,329.92 kWh/year.

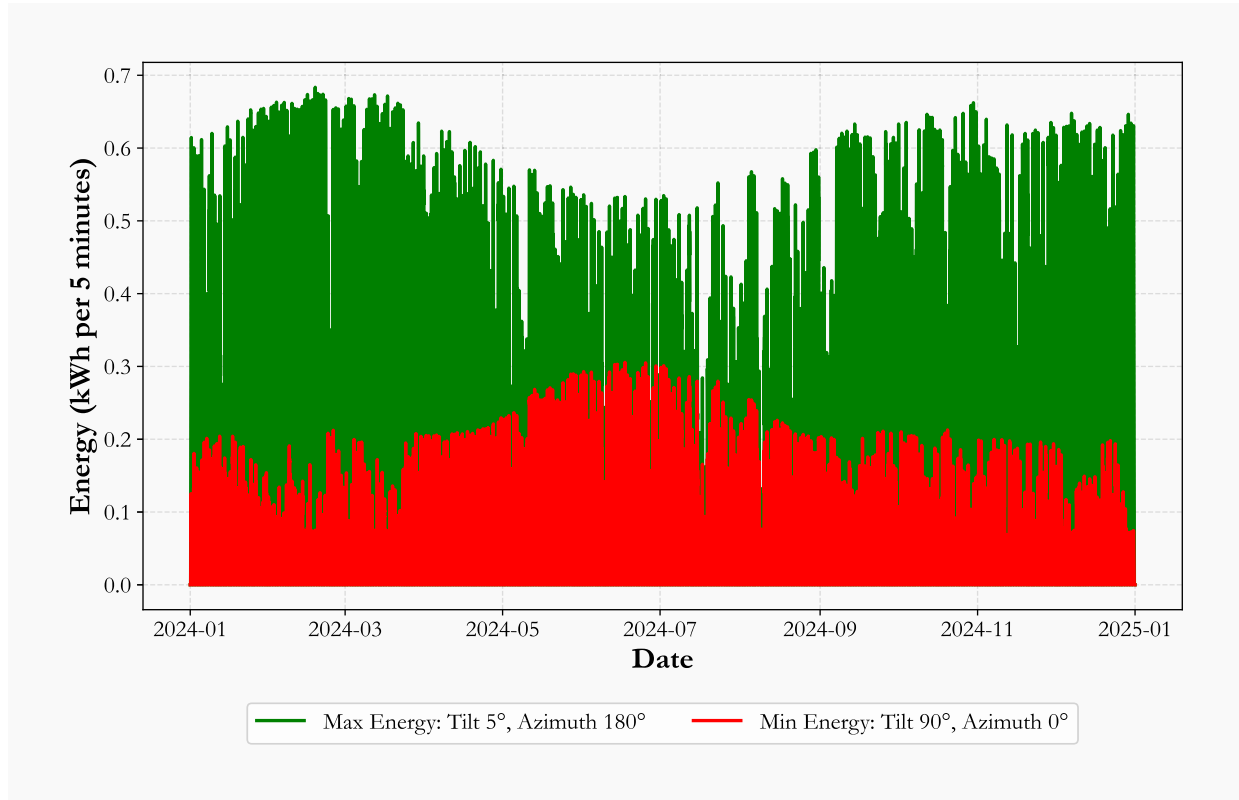


FIGURE 36. Annual 2024 energy at Njagu-Mhasibu-Ruiru Estate, Kenya (5-min resolution): maximum of 14,911.01 kWh at 5°–180° and minimum of 5,294.46 kWh at 90°–0° (tilt–azimuth). Optimal fixed setup: 5° tilt facing true south (180° azimuth).

zooming in on small time periods, e.g., sub hourly or weekly plots, in order to clearly observe the variations without limitations by screen pixels as seen in the energy graphs 14 and 15.

F. SEASONAL VARIABILITY IN SOLAR PV ENERGY OUTPUT

The solar energy generation profile for Njagu, Ruiru, exhibits significant seasonal variations, as depicted in Figure 31. The months of February and March demonstrate consistent and high solar energy production, which can be attributed to relatively clear sky conditions and high solar irradiance levels. In contrast, the period spanning April to August 2024, energy generation is lower and more erratic, reflecting increased cloud cover and seasonal rainfall.

G. DIURNAL SOLAR PV ENERGY OUTPUT TRENDS

The system's daily energy profile follows a well-defined diurnal pattern, as portrayed in Figure 31, with peak production occurring around solar noon and declining to zero during night hours. Intra-day fluctuations are likely due to intermittent cloud shading effects, which cause transient dips in power generation. These high-frequency power variations, as seen in Figure 33, could pose challenges for grid integration and require mitigation strategies such as battery storage or predictive control mechanisms.

H. GHI VS MODULE TEMPERATURE

The module temperatures varied between 41.8 °C and 51.4 °C, despite GHI values fluctuating significantly from 12.9 W/m² to 350.3 W/m² as shown in Figure 37. In Figure 38, the module temperatures range from 43.3 °C to 49.1 °C with GHI between 26.4 W/m² and 397.2 W/m². This apparent decoupling between short-term irradiance variability and module temperature suggests the influence of thermal inertia in the system. Since solar PV cell efficiency declines with temperature (typically 0.4–0.5% per °C above 25 °C), a system with lower thermal inertia may experience more frequent and severe efficiency losses during short irradiance bursts. Thermal inertia helps reduce these short-term efficiency dips, improving the average performance over time.

VI. ERROR METRICS: MODEL EVALUATION AND COMPARATIVE ANALYSIS

To validate the OSM-MEPS model within the scope of energy capture and conversion—from solar irradiance to inverter AC output—three benchmark tools were selected for their complementary strengths: PVLIB provides a physics-based, component-level reference for validating solar geometry, POA irradiance, module temperature and DC/AC power calculations [69]. The System Advisor Model (SAM) offers a system-level performance benchmark, incorporating practical losses such as shading, soiling and inverter clipping,

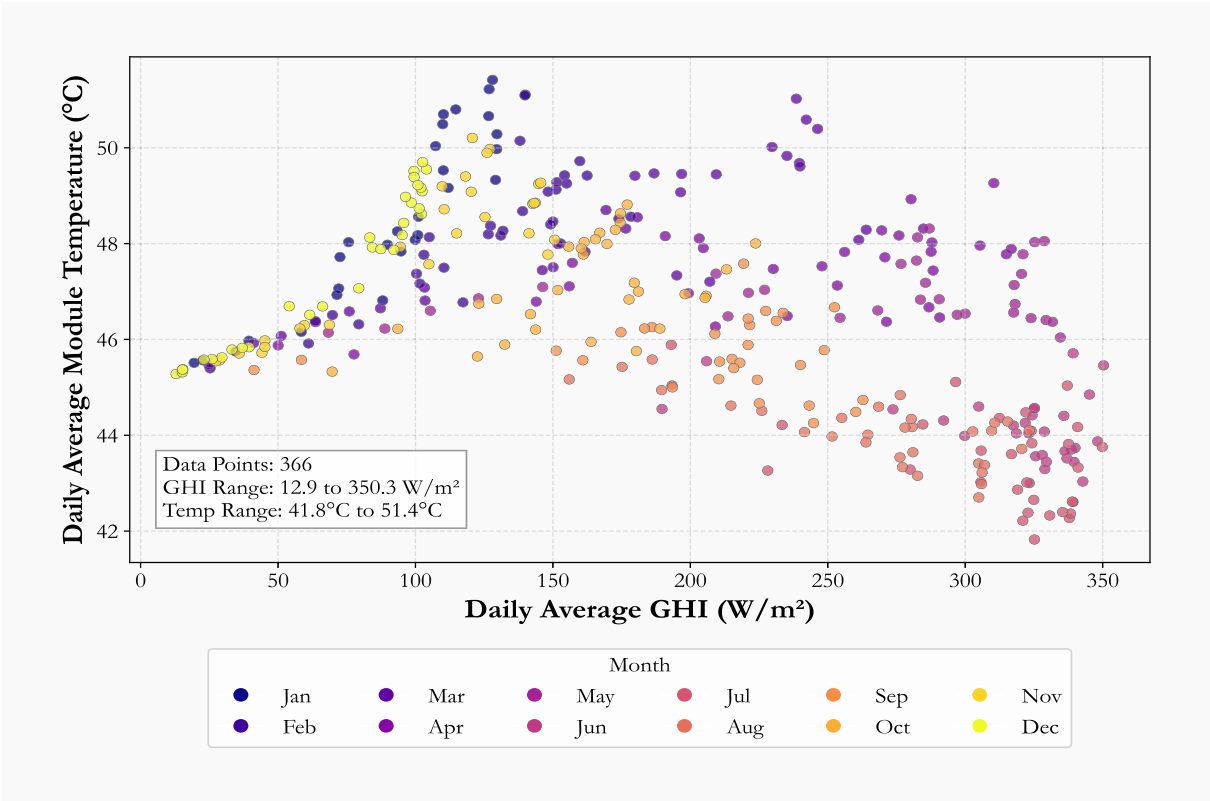


FIGURE 37. GHI vs module temperature for Serres C solar PV plant, Greece.

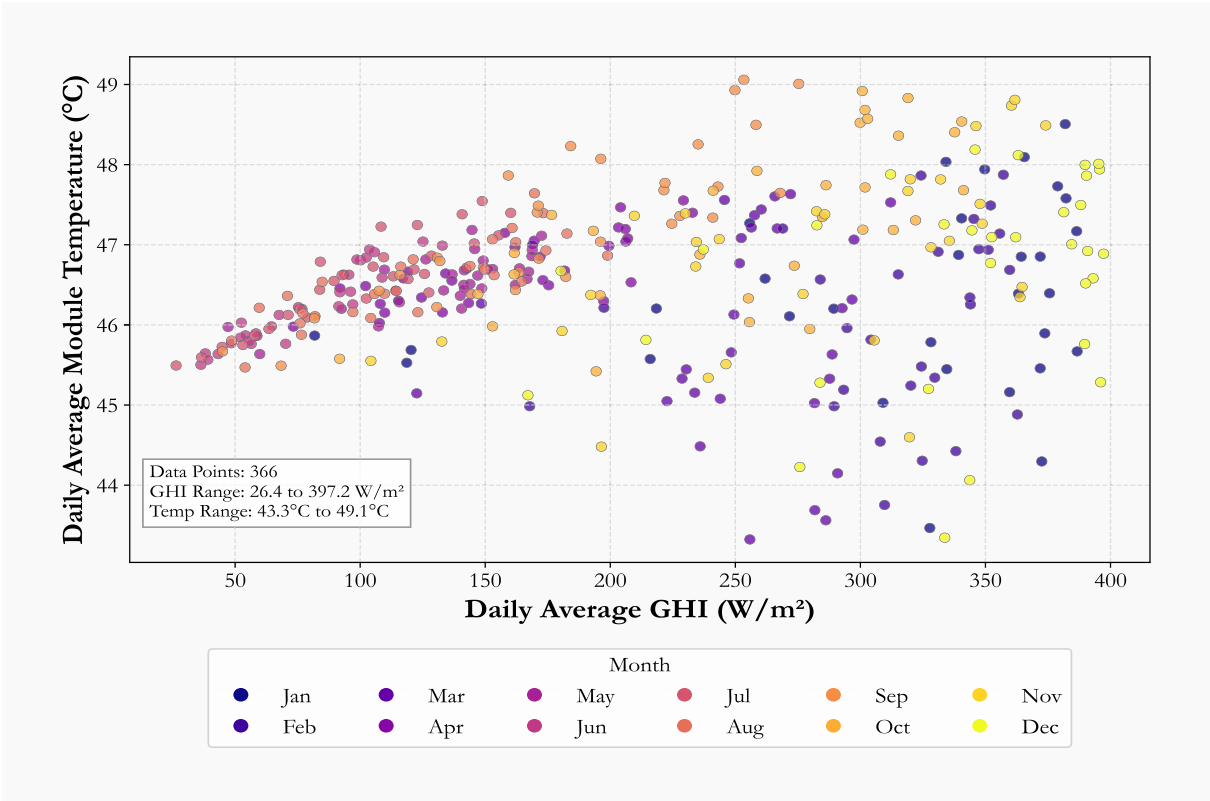


FIGURE 38. GHI vs module temperature for 367 Glen Osmond rooftop solar PV, Adelaide, Australia.

reflecting real-world behavior [13]. The GSA serves as a satellite-derived, location-specific benchmark for annual energy yield, supporting geospatial validation [36]. Together, these tools create a robust triangulation strategy to evaluate the technical accuracy and generalizability of the OSM-MEPS model, excluding financial modeling aspects of solar PV systems, which are beyond this study's scope.

An evaluation of the PVLIB and OSM-MEPS models was conducted using empirical solar PV energy output data collected across diverse geographic locations. To quantify the models' predictive accuracy, three standard statistical error metrics were employed: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). These metrics were computed by comparing each model's output against actual energy generation measurements obtained from PVOutput [76] for each site.

The MAE measures the average magnitude of prediction errors, providing a linear sense of deviation regardless of direction:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (51)$$

The RMSE penalizes larger errors more heavily by squaring the residuals before averaging:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (52)$$

The coefficient of determination R^2 evaluates how well the predicted values capture the variance in the observed data, with values ranging from 0 to 1:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (53)$$

where y_i is the observed value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the mean of the observed values and n is the number of time steps.

Both models utilized high-resolution irradiance and weather input from Solcast-DNV. According to independent validation studies, Solcast's data demonstrated a strong agreement with ground-based measurements, yielding a normalized Mean Bias Error (nMBE) of 2.65%, normalized Mean Absolute Error (nMAE) of 10.59%, normalized RMSE (nRMSE) of 15.43%, and an R^2 of 0.98 [77]. Solcast's methodology integrates first-principles physics (Beer-Lambert law, Rayleigh and Mie scattering), satellite-derived cloud motion tracking, aerosol and water vapor modeling, machine learning enhancements for splitting GHI into DNI and DHI and cloud-edge irradiance corrections [77].

Statistical evaluation, summarized in Table 8, shows that the OSM-MEPS model closely matches the actual energy output measured at the Serres-C site in Greece, achieving a mean absolute error (MAE) of 469.57 kWh, a root mean squared error (RMSE) of 584.62 kWh and an R^2 value of 0.887 over the full year. In comparison, PVLIB recorded a

MAE of 1,400.39 kWh, RMSE of 1,526.11 kWh and a weaker R^2 of 0.232. Seasonal tests further confirmed OSM-MEPS's robustness. In May 2024, a month of high solar variability, the novel model achieved an R^2 of 0.917, vastly outperforming PVLIB's -0.039, with a fourfold lower MAE (390.07 kWh vs. 1,586.84 kWh). In December 2024, the OSM-MEPS model continued to outperform PVLIB, with a MAE of 380.02 kWh, RMSE of 490.17 kWh and R^2 of 0.873, compared to PVLIB's MAE of 674.01 kWh, RMSE of 826.40 kWh and R^2 of 0.638.

Geographic tests showed similar trends. In Pretoria, the OSM-MEPS model yielded a MAE of 6.33 kWh and R^2 of 0.443, while PVLIB returned a negative R^2 of -0.239. In Adelaide, PVLIB failed entirely (R^2 = -0.959, MAE = 23.96 kWh and RMSE = 26.27), whereas the OSM-MEPS model remained reliable (R^2 = 0.866, MAE = 5.16 kWh and RMSE = 6.86). In Kisumu, both models showed similar MAE (0.88 kWh) but weak correlations (PVLIB: R^2 = 0.127; OSM-MEPS: R^2 = 0.051). These errors were influenced by data quality limitations, such as the absence of complete annual records and data logging issues reported on the PVOutput platform [76]. Furthermore, medium shading was noted in the metadata but lacked temporal details or intensity profiles, limiting the ability to model its effects. Additional factors such as unreported soiling, maintenance downtime, and atmospheric smoke from burning refuse may have also contributed to uncertainty in the model outputs.

The results highlight OSM-MEPS's superior accuracy in capturing daily and seasonal dynamics across diverse climates. Its performance edge is most evident during high-variability months and in non-temperate regions, demonstrating geographic generalizability. While PVLIB remains accessible for basic modeling, its deterministic structure and lack of integrated empirical loss adjustments limit its forecasting reliability. These findings underscore the importance of thermal coupling in solar PV module design, advanced irradiance modeling and loss characterization for improving photovoltaic output simulations.

A. MODEL FRAMEWORK ANALYSIS: OSM-MEPS VS. PVLIB VS. GSA VS. SAM

OSM-MEPS model *cloud opacity adjustment* and *dynamic albedo* outperform GSA's static albedo and SAM's dependence on user-defined values. A comparison of the POA irradiance models is provided in Table 3. The NOCT based temperature model in OSM-MEPS model, initialized with a nominal value of 45 °C (see Table 4), provides a generalized and physically grounded alternative to the Sandia Array Performance Model (SAPM). This links irradiance and module temperature using a linear scaling relationship anchored in standard test conditions (STC), avoiding the regional overfitting issues associated with SAPM, originally tuned using data from hot, arid environments.

SAPM includes an exponential wind-dependent cooling term, $e^{b \times \text{wind}}$, intended to model convective heat transfer. However, under low wind conditions (wind ≈ 0), this

TABLE 3. Comparison of POA irradiance models across the frameworks [13].

Model	Direct Irradiance	Diffuse Irradiance	Albedo	Mid Latitude Performance
OSM-MEPS	$dni \times \cos(AOI) \times \left(1 - \frac{\text{cloud opacity}}{100}\right)$	Isotropic: $\frac{1 + \cos(\text{tilt})}{2}$	Dynamic albedo (0.1–0.9)	Explicit mid-latitude adaptations
PVLib	$dni \times \cos(AOI)$	Perez/Hay-Davies (anisotropic) + isotropic	Fixed (0.25)	Mid-latitude bias
GSA	Simplified clear-sky model	ERA5 reanalysis data	Fixed (0.2)	Low-resolution (1 km), ignores frost albedo
SAM	DISC/BRIGHT models	Perez/Hay-Davies	Fixed or user-defined	Customizable but complex

TABLE 4. Temperature modeling approaches in PV frameworks [13].

Model	Temperature Formula	Wind Cooling	Mid Latitude Adaptations
OSM-MEPS	$45^\circ \text{C} + \frac{\text{POA}}{1000} \times (28 - T_{\text{air}})$	Implicit in $(28 - T_{\text{air}})$	Optimized for cold
PVLib (SAPM)	$T_{\text{air}} + a \times \text{POA} \times e^{b \times \text{wind}}$	Explicit ($b = -0.075$)	Overcools in low wind
GSA	N/A (energy-only)	N/A	N/A
SAM	NOCT or Sandia models	Configurable	Requires manual tuning

simplifies to:

$$T_{\text{mod}} \approx T_{\text{air}} + a \times \text{POA} \quad (54)$$

which tends to underestimate actual module temperature due to overestimated cooling. This results in a systematic *overcooling bias*, particularly problematic in wind-sheltered or mid-latitude rooftop solar PV systems.

In summary, the OSM-MEPS model overcomes the wind sensitivity of PVLIB (SAPM) and the configuration complexity of SAM, resulting in more robust, consistent and climate-adaptive performance predictions across a wide range of geographic conditions.

The OSM-MEPS model's linear scaling avoids the single-diode inaccuracies encountered in polar twilight conditions, whereas SAM requires manual adjustments to address these issues. The GSA offers free, globally accessible data, making it well-suited for an initial site survey. However, it lacks support for open-source module temperature modeling and detailed physical representations, limiting its precision for in-depth performance analysis. The System Advisor Model (SAM) stands out for its extensive configurability, including features such

TABLE 5. DC/AC Power conversion and polar performance comparison [13].

Model	DC Power Model	Low-Light Handling	Mid Latitude Losses
OSM-MEPS	Linear scaling $\left(\frac{\text{POA}}{\text{STC}}\right) + \text{temperature correction}$	Robust ($< 200 \text{ W/m}^2$)	Relative Humidity ($1 - 0.002 \times RH$)
PVLib	Single-diode model	Fails at low irradiance	Generic losses
GSA	Not disclosed	N/A	N/A
SAM	Single-diode or empirical	User-defined thresholds	Configurable

TABLE 6. Computational performance summary: OSM-MEPS Model + PVLIB + GSA + SERRES-C PV energy output data.

Metric	Value
Execution Time	36.67 seconds
Peak Memory Usage	79.80 MB
Current Memory Usage	61.00 MB
Data Resolution	5-minute intervals
Time Span Simulated	1 year (2024)
Hardware Used	Intel® Core™ i7 - 4790 CPU @ 3.60GHz (4 Cores, 8 Threads), 16 GB RAM.

TABLE 7. Integration of key frameworks in OSM-MEPS.

Framework	Core Contribution	How OSM-MEPS Integrates the Framework
The Scientific Method	Hypothesis testing and empirical validation.	Ensures models are grounded in real-world data (e.g., Solcast inputs, Serres-C PV plant outputs).
Engineering Design Process	Practical problem-solving and iterative refinement.	Focuses on pragmatic and iterative approaches in problem solving (e.g., Kenya's rooftop solar PV energy viability analysis).
First-Principles Thinking	Fundamental physics-based modeling.	Builds models from basic laws (e.g., solar irradiance physics) to avoid oversimplification.

as bifacial and tracking systems. Despite this flexibility, it presents a steep learning curve and requires expert knowledge to accurately model cold or polar conditions. To enhance its accuracy and usability, it is recommended to integrate high-resolution, error-minimized, and validated data.

TABLE 8. Solar PV model performance comparison in select locations: North and south hemispheres.

LOCATION	PERIOD	MODEL	MAE (kWh)	RMSE (kWh)	R ²
Serres C, Greece	Dec 2024	PVLIB	674.01	826.40	0.638
Serres C, Greece	Dec 2024	OSM-MEPS	380.02	490.17	0.873
Serres C, Greece	May 2024	PVLIB	1586.84	1741.99	−0.039
Serres C, Greece	May 2024	OSM-MEPS	390.07	492.00	0.917
Serres C, Greece	2024	PVLIB	1400.39	1526.11	0.232
Serres C, Greece	2024	OSM-MEPS	469.57	584.62	0.887
Pretoria, South Africa	2024	PVLIB	10.44	11.75	−0.239
Pretoria, South Africa	2024	OSM-MEPS	6.33	7.88	0.443
Adelaide, Australia	2024	PVLIB	23.96	26.27	−0.959
Adelaide, Australia	2024	OSM-MEPS	5.16	6.86	0.866
Kisumu, Kenya	2024	PVLIB	0.88	–	0.127
Kisumu, Kenya	2024	OSM-MEPS	0.88	–	0.051

Notes: MAE = Mean Absolute Error, RMSE = Root Mean Square Error. Dashes indicate unavailable metrics. RMSE is not computable if the predicted or actual data arrays are empty (missing empirical/site-measured data in Kisumu, Kenya). All values are rounded to two decimal places. The OSM-MEPS model demonstrates superior performance across all locations with 43–75% lower MAE and consistently higher R² values.

B. OSM-MEPS MODEL COMPUTATIONAL DEMANDS AND PRACTICAL SCALABILITY

The OSM-MEPS model demonstrates practical scalability and real-world feasibility by balancing physical accuracy, computational efficiency and geographic adaptability. For comparative validation, the model was implemented with open-source libraries such as `pvlb`, `NumPy` and `pandas`. This simulation was performed on an HP Z230 Tower Workstation, with hardware specifications outlined in Table 6. Using the full 2024 input data at 5-minute resolution—including weather and SERRES-C solar PV site measured energy—the simulation, run in a Python environment, quantified energy output and compared results across the OSM-MEPS model, PVLIB, and GSA, completing in approximately 36.67 seconds with peak memory usage below 80 MB.

1) HORIZONTAL SCALABILITY

The OSM-MEPS model is developed in Python, a widely used language in scientific computing due to its simplicity, flexibility and extensive support for parallelism. It leverages Python's *multiprocessing* (via `multiprocessing.Pool`) and *asynchronous I/O* capabilities (`asyncio` or `aiohttp`), which allow efficient management and execution of multiple simulations concurrently without blocking system resources. This enables batch processing of solar PV energy simulations across multiple geographic sites in parallel, significantly improving computational speed and resource utilization.

By supporting simultaneous simulations of diverse PV installations—varying in location, orientation and system configuration—the OSM-MEPS model is highly suited for large-scale, multi-location studies, including national PV deployment planning and climate-sensitivity assessments. Despite the expanded scale, the model maintains a consistent level of physical accuracy, ensuring performance metrics and irradiance estimates remain reliable across all sites. This design balances computational scalability with precision,

making it applicable to both research and real-world grid integration analysis.

2) VERTICAL SCALABILITY

Python's modular architecture enables OSM-MEPS to accommodate a wide range of photovoltaic (PV) system scales—from small, residential rooftop arrays to large, utility-scale solar farms—with minimal structural changes to the core model. This is achieved through the use of reusable functions, scalable input parameters and object-oriented design principles that allow system components (e.g., inverter capacity, panel configurations, irradiance inputs) to be dynamically adjusted without rewriting the modeling logic. Such flexibility is essential for adapting the same modeling framework to different deployment contexts, including urban microgrids, rural electrification, commercial rooftops and grid-connected utility plants. Users can easily modify parameters such as array size, orientation, shading losses or location-specific weather inputs to reflect the physical characteristics of each site. This scalability ensures that OSM-MEPS model remains robust and relevant across varying project sizes, enabling comparative analysis, policy evaluation and tailored design optimization for specific operational goals or geographic conditions.

3) DATA COMPATIBILITY AND PREPROCESSING

The OSM-MEPS model demonstrates strong compatibility with diverse meteorological datasets, requiring only minimal adaptation, such as renaming columns or metric units alignment, to integrate widely used formats e.g., CSV and Excel from sources such as Solargis and Solcast. This flexibility is supported by Python's robust data handling ecosystem: `pandas` enables structured data manipulation, `xarray` excels at processing multi-dimensional meteorological variables like temperature and irradiance, `NumPy` provides efficient numerical operations, `openpyxl` facilitates Excel file handling and `pvlb` delivers solar-specific

Algorithm 1 Solar Panel Tilt and Azimuth Optimization for PV Energy

Input: Weather CSV file with `period_end`, `dni`, `ghi`, `dhi`, `air_temp`, `albedo`, `zenith`, `azimuth`, `relative_humidity`, `cloud_opacity`

Output: Optimal and worst-case tilt-azimuth angles with corresponding total energy (kWh)

- 1: Load and pre-process CSV data (timestamp conversion and indexing)
- 2: Validate and fill/interpolate missing required weather columns
- 3: Define system parameters: panel rating, panel area, total rooftop area
- 4: Compute number of panels based on total and unit area
- 5: Define tilt angle range (0° — 91° , step = 5°)
- 6: Define azimuth angle range (0° — 360° , step = 10°)
- 7: Initialize tracking variables for highest and lowest energy output
- 8: **for** each candidate tilt angle **do**
- 9: **for** each candidate azimuth angle **do**
- 10: Convert input angles and solar angles to radians
- 11: Calculate angle of incidence (AOI)
- 12: Compute direct irradiance on tilted surface (`poa_direct`)
- 13: Compute sky diffuse irradiance (`poa_diffuse`)
- 14: Compute ground-reflected irradiance (`poa_sky_diffuse`)
- 15: Sum irradiance components to get total POA irradiance (`poa_total`)
- 16: Estimate module temperature (`module_temp`)
- 17: Adjust panel power output for temperature (`panel_power`)
- 18: Compute DC power output (`dc_power`)
- 19: Convert DC power to AC power using inverter efficiency (`ac_power`)
- 20: Scale AC power by number of panels (`scaled_power`)
- 21: Compute energy for each interval in kWh (`energy_kwh`)
- 22: Aggregate total energy for the configuration
- 23: **if** total energy > current maximum **then**
- 24: Update best tilt/azimuth and store energy profile
- 25: **end if**
- 26: **if** total energy < current minimum **then**
- 27: Update worst tilt/azimuth and store energy profile
- 28: **end if**
- 29: **end for**
- 30: **end for**
- 31: Output optimal and worst tilt/azimuth and their energy yields

computational tools. Together, these libraries streamline preprocessing tasks including data cleaning, variable mapping, time index alignment and temporal resampling (e.g., from five-minute to hourly intervals). As a result, the OSM-MEPS model efficiently integrates high-resolution input data while preserving integrity and reproducibility. This design minimizes preprocessing effort, enhances scalability and supports rapid deployment across varied solar PV modeling applications.

4) SYSTEM INTEGRATION SCALABILITY

Python's modular architecture allows the OSM-MEPS model to interface smoothly with larger energy modeling frameworks. This enables its outputs—such as high-resolution

solar PV generation profiles—to feed directly into grid-level simulations assessing stability, flexibility and ramping needs. As solar PV penetration increases, accurately capturing variability and intermittency becomes critical. The OSM-MEPS model supports scalable integration by offering real-time or historical irradiance data that can be used to test system resilience, optimize dispatch strategies and evaluate ancillary services under diverse operational scenarios.

5) OPERATIONAL ROBUSTNESS

Python's ecosystem provides built-in tools for handling missing data, validating model outputs against empirical datasets such as [76] and exporting results into formats compatible with grid simulation frameworks. These features ensure that the OSM-MEPS model remains robust, transparent and ready for deployment in both academic and professional settings.

C. OSM-MEPS HYBRID FRAMEWORK CONTRIBUTION

OSM-MEPS method addresses a theoretical gap in the modeling frameworks by proactively integrating first-principles thinking, ensuring a review of underlying assumptions in modeling and providing fresh insights 7.

The OSM-MEPS overarching framework addresses key gaps in energy and power systems modeling by integrating first-principles, scientific inquiry and iterative engineering design. It ensures rigor through transparent physics-based equations and systematic validation with real-world data, effectively bridging theory and practical deployment for enhanced credibility and applicability. This iterative philosophy fosters continuous refinement through testing and validation: future enhancements include high-resolution terrain mapping, urban and microclimate modeling, dynamic 3D shading analysis, panel-specific degradation and soiling models and system outage accounting with the aim of minimizing discrepancies, as seen at the Victoria-Views site in Kisumu, Kenya (see Figure 28), between the physics modeled and actual Solar PV plant energy outputs.

VII. CONCLUSION

The OSM-MEPS method represents a novel modeling framework systematically derived and synthesized from energy and power systems modeling literature, integrating first-principles physics, the scientific method, and iterative engineering design. It addresses key limitations in existing tools like PVLIB and GSA by enhancing model accuracy across diverse regions. By grounding the analysis in fundamental physics and domain knowledge—such as the impact of cloud cover on solar radiation—the method emphasizes real causal factors rather than mere data fitting. The scientific method tests hypotheses and accounts for external influences to better understand how tilt angles and weather affect solar energy capture. The engineering design approach incorporates iterative testing and refinement, ensuring sustained accuracy across varying conditions.

The resulting OSM-MEPS model utilizes high-resolution, key meteorological inputs—such as DNI, GHI, DHI,

temperature, relative humidity and cloud opacity—to simulate photovoltaic energy output. It has been validated with actual measurements and compared with existing models, showing improved accuracy in short-term and yearly energy estimates. The model captures seasonal changes well, even across diverse climates. Due to its modular Python implementation, the model can readily integrate into comprehensive power system flexibility workflows, facilitating the evaluation of grid stability, reserve requirements and ramping challenges in scenarios with high solar PV energy penetration.

VIII. RECOMMENDATION

Future research should incorporate localized, high-resolution sub-minute weather data. Advances in sensing technologies and computational resources will make this feasible. The goal is to reduce forecasting errors. Extending the OSM-MEPS framework to model advanced PV array configurations (e.g., dual-axis tracking, terrain-aware shading, bifacial modules) will enhance fidelity, allowing more accurate representation of PV dynamics across varied geographies. Also, integrating time-resolved load data that reflects real-world consumption behaviors through human-centered design will further strengthen simulation accuracy and inform effective demand-response strategies in solar PV integrated grids.

Additionally, hybridizing OSM-MEPS with physics-informed neural networks (PINNs) offers a promising pathway to combine first-principles rigor with data-driven adaptability, particularly valuable where empirical data remains limited. Incorporating real-time forecasting with adaptive calibration would enable the model to dynamically adjust to live measurements—such as sudden cloud cover or temperature changes. Collectively, these enhancements will improve computational efficiency and practical relevance, supporting grid operators in managing the variability of solar PV generation more effectively.

APPENDIX

PROGRAMMING AND MODELING CODE

All source code, data files and reproducible scripts used in this study are openly available in the GitHub repository: `Github: OSM-MEPS Model Python code`. Site-measured solar PV output data is accessible via the PVOutput online platform: `Serres-C Solar PV Plant: PVOutput`. Solar resource assessment and site-specific data were also sourced from the Solargis (GSA) web platform for:

- Njagu-Mhasibu-Ruiru Estate, Kenya,
- Serres-C Solar PV Plant, Greece.

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